Eigenvalues and Quasirandom Hypergraphs

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Abstract

Let $p(k)$ denote the partition function of k. For each $k \geq 2$, we describe a list of $p(k) - 1$ quasirandom properties that a k-uniform hypergraph can have. Our work connects previous notions on hypergraph quasirandomness, beginning with the early work of Chung and Graham and Frankl-Rödl related to strong hypergraph regularity, the spectral approach of Friedman-Wigderson, and more recent results of Kohayakawa-Rödl-Skokan and Conlon-Hàn-Person-Schacht on weak hypergraph regularity and its relation to counting linear hypergraphs.

For each of the quasirandom properties that are described, we define a hypergraph eigenvalue analogous to the graph case and a hypergraph extension of a graph cycle of even length whose count determines if a hypergraph satisfies the property. This answers a question of Conlon et al. Our work can be viewed as an extension to hypergraphs of the seminal results of Chung-Graham-Wilson for graphs.

1 Introduction

The study of quasirandom or pseudorandom graphs was initiated by Thomason [50, 51] and then refined by Chung, Graham, and Wilson [15], resulting in a list of equivalent (deterministic) properties of graph sequences which are inspired by $G(n, p)$. Beginning with these foundational papers on the subject [15, 50, 51], the last two decades have seen an explosive growth in the study of quasirandom structures in mathematics and computer science. For details on quasirandomness, we refer the reader to a survey of Krivelevich and Sudakov [34] for graphs and recent papers of Gowers [25, 26, 27] for general quasirandom structures including hypergraphs.

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1.1 Previous Results

The core of what Chung, Graham, and Wilson [15] proved is the following remarkable theorem. We write $o(n^2)$ for a function $f : \mathbb{N} \to \mathbb{R}$ such that $f(n)n^{-2} \to 0$ as $n \to \infty$. The precise way to interpret the $o(\cdot)$ terms in the following theorems is explained in the discussion after Theorem 3. As usual, for a graph or hypergraph G, we write $V(G)$ for its vertex set and $E(G)$ for its edge set. Unless specifically mentioned, all graphs and hypergraphs in this paper have no loops or multiple edges. For $U \subseteq V(G)$, the induced subgraph on U, denoted $G[U]$, is the graph with vertex set U and edge set $\{e \in E(G) : e \subseteq U\}$. If F and G are graphs or hypergraphs, a *labeled copy of F in G* is an edge-preserving injection $V(F) \to V(G)$, i.e. an injection $\alpha : V(F) \to V(G)$ such that if E is an edge of F, then $\{\alpha(x) : x \in E\}$ is an edge of G.

Theorem 1. *(**Chung-Graham-Wilson [15]**) Let $0 < p < 1$ be a fixed constant and let $\mathcal{G} = \{G_n\}_{n\to\infty}$ be a sequence of graphs such that $|V(G_n)| = n$ and $|E(G_n)| \geq p\binom{n}{2} + o(n^2)$. The following properties are equivalent:*

- `Disc`: *(short for discrepency) For every $U \subseteq V(G_n)$, $|E(G_n[U])| = p\binom{|U|}{2} + o(n^2)$.*
- `Expand`: *For every $S, T \subseteq V(G_n)$, $e(S,T) = p|S||T| + o(n^2)$, where $e(S,T)$ is the number of edges with one endpoint in S and one endpoint in T, with edges inside $S \cap T$ counted twice.*
- `Count[All]`: *For every fixed graph F with f vertices and e edges, the number of labeled copies of F in G_n is $p^e n^f + o(n^f)$.*
- $\texttt{Cycle}_4$: *The number of labeled copies of C_4 in G_n is at most $p^4 n^4 + o(n^4)$.*
- $\texttt{Cycle}_{2\ell}$: *For $\ell \geq 2$ a fixed integer, the number of labeled copies of $C_{2\ell}$ in G_n is at most $p^{2\ell} n^{2\ell} + o(n^{2\ell})$.*
- `Eig`: *If λ_1 and λ_2 are the first and second largest eigenvalues in absolute value of the adjacency matrix of G_n respectively, then $|\lambda_1| = pn + o(n)$ and $|\lambda_2| = o(n)$.*

Chung, Graham, and Wilson [15] proved several more properties, but the five properties stated in Theorem 1 are usually viewed as central to the field. In particular, `Disc` and `Expand` are basic properties of random graphs. Note that Chung, Graham, and Wilson [15] put the condition "$|E(G_n)| \geq p\binom{n}{2} + o(n^2)$" into the statement of the properties, but since it appears in every property the condition can be moved to a requirement on the graph sequence as in Theorem 1.

Almost immediately after proving Theorem 1, Chung and Graham [8, 9, 12, 13, 14] investigated generalizing the theorem to k-uniform hypergraphs. A *k-uniform hypergraph* is a pair of finite sets $(V(H), E(H))$ where $E(H) \subseteq \binom{V(H)}{k}$ is a collection of k-subsets of $V(H)$. One initial difficulty in generalizing Theorem 1 to $k > 2$ is a construction of Erdős and Hajnal [20].

Construction. (Erdős and Hajnal [20]) Consider the three-uniform hypergraph formed as follows (actually, a sequence of probability spaces of n vertex hypergraphs, one for each integer n). Let T_n be a random (graph) tournament on n vertices, by which we mean an orientation of the edges of the complete graph K_n where each edge is oriented in one of two directions with equal probability and the orientations are done independently for distinct edges. Form an n-vertex, three-uniform hypergraph H_n by letting three vertices u, v, w form an edge of H_n if $u \to v \to w \to u$.

Rödl observed that this construction shows that the hypergraph generalizations of `Disc` and `Count[All]` are not equivalent. The sequence H_n is easily seen to satisfy `Disc`, since in any $U \subseteq V(H_n)$ the probability that three vertices from U form an edge of H_n is $p = \frac{1}{4}$, the same as the expected density of H_n. On the other hand, it is easily observed that H_n does not contain any copy of $K_4^{(3)}$, the complete three-uniform hypergraph on four vertices. Since the number of $K_4^{(3)}$ in H_n is zero, the graph sequence H_n obviously fails `Count[All]`. Motivated by this, Chung and Graham [9, 12, 13, 14] investigated how to strengthen the properties `Disc`/`Expand` to make them equivalent to `Count[All]`. They found several properties equivalent to `Count[All]`; the main property they use is related to the number of even/odd subgraphs of a given hypergraph which they called `Deviation`. Subsequently, properties equivalent to `Count[All]` have been studied by several researchers [6, 25, 31, 33].

The above work provides many properties equivalent to `Count[All]` in general hypergraphs, but it remained open whether the simpler property `Disc`/`Expand` for k-uniform hypergraphs is equivalent to counting some class of hypergraphs or counting a single substructure. This is related to the Weak Hypergraph Regularity Lemma [10, 22, 46]. Recently, Kohayakawa, Nagle, Rödl, and Schacht [32] answered this question by showing that `Disc` is equivalent to counting the family of linear hypergraphs, where a hypergraph H is *linear* if for every $E \neq E' \in E(H)$, then $|E \cap E'| \leq 1$. Building on this, Conlon, Hàn, Person, and Schacht [17] showed that `Disc` is equivalent to counting a type of linear four cycle. These two results can be combined into the following theorem.

Theorem 2. ***(Kohayakawa-Nagle-Rödl-Schacht [32] and Conlon-Hàn-Person-Schacht [17])*** *Let $0 < p < 1$ be a fixed constant and let $\mathcal{H} = \{H_n\}_{n\to\infty}$ be a sequence of k-uniform hypergraphs such that $|V(H_n)| = n$ and $|E(H_n)| \geq p\binom{n}{k} + o(n^k)$. The following properties are equivalent:*

- `Disc`*: For every $U \subseteq V(H_n)$, $|E(H_n[U])| = p\binom{|U|}{k} + o(n^k)$.*
- `Count[linear]`*: For every fixed linear k-uniform hypergraph F with e edges and f vertices, the number of labeled copies of F in H_n is $p^e n^f + o(n^f)$.*
- `Cycle`$_4$*: The number of labeled copies of C_4 in H_n is at most $p^{|E(C_4)|}n^{|V(C_4)|} + o(n^{|V(C_4)|})$, where C_4 is a linear hypergraph defined precisely in Section 2.*

Note that Conlon et al. [17] put the condition "$|E(H_n)| \geq p\binom{n}{k} + o(n^k)$" into the statement of the properties that don't trivially imply it like `Disc` and this is equivalent to the way we have stated Theorem 2. Conlon et al. [17] have several more properties including induced subgraph counts and common neighborhood sizes, but we consider the properties stated in Theorem 2 as the core properties. Theorem 2 equates `Disc`, an exponential time checkable property, with $\texttt{Cycle}_4$, a property which can be checked in polynomial time.

1.2 Our Results

Neither Chung and Graham [9, 12, 13, 14] nor Kohayakawa, Rödl, and Skokan [33] provided a generalization of `Eig` to hypergraphs. Later, Conlon, Hàn, Person, and Schacht [17] asked whether there exists a generalization of `Eig` to k-uniform hypergraphs. For graphs, `Eig` helps enormously to simplify the proof that $\texttt{Count}[C_4] \Rightarrow \texttt{Expand}$. In addition, a spectral characterization of quasirandomness has proved to be a very useful result in many situations, for example in the proofs that certain explicitly constructed graphs are quasirandom (see [3, 4, 36, 48]).

This leads to our first main contribution. We define a generalization of `Eig` to k-uniform hypergraphs and add it into the equivalences stated in Theorem 2. This facilitates the proof of the implication $\texttt{Cycle}_4 \Rightarrow \texttt{Disc}$ and also aids the development of spectral methods of quasirandom hypergraphs, a direction that has proved very fruitful for graphs.

Our second main contribution is to generalize Theorem 2 substantially. Let $k \geq 2$ be an integer and let π be a proper partition of k, by which we mean that π is an unordered list of at least two positive integers whose sum is k. For the partition π of k given by $k = k_1 + \cdots + k_t$, we will abuse notation by saying that $\pi = k_1 + \cdots + k_t$. For every proper partition π, we define properties $\texttt{Expand}[\pi]$, $\texttt{Eig}[\pi]$, and $\texttt{Cycle}_4[\pi]$ and show that they are equivalent.

Definition. *Let $k \geq 2$ and let $\pi = k_1 + \cdots + k_t$ be a proper partition of k. A k-uniform hypergraph F is* π-linear *if there exists an ordering $E_1, \ldots, E_m$ of the edges of F such that for every i, there exists a partition of the vertices of E_i into $A_{i,1}, \ldots, A_{i,t}$ such that for $1 \leq s \leq t$, $|A_{i,s}| = k_s$ and for every $j < i$, there exists an s such that $E_j \cap E_i \subseteq A_{i,s}$.*

The following is our main theorem.

Theorem 3. ***(Main Result)*** *Let $0 < p < 1$ be a fixed constant and let $\mathcal{H} = \{H_n\}_{n\to\infty}$ be a sequence of k-uniform hypergraphs such that $|V(H_n)| = n$ and $|E(H_n)| \geq p\binom{n}{k} + o(n^k)$. Let $\pi = k_1 + \cdots + k_t$ be a proper partition of k. The following properties are equivalent:*

- ***Eig**[π]: $\lambda_{1,\pi}(H_n) = pn^{k/2} + o(n^{k/2})$ and $\lambda_{2,\pi}(H_n) = o(n^{k/2})$, where $\lambda_{1,\pi}(H_n)$ and $\lambda_{2,\pi}(H_n)$ are the first and second largest eigenvalues of H_n with respect to π, which are defined in Section 3.*
- ***Expand**[π]: For all $S_i \subseteq \binom{V(H_n)}{k_i}$ where $1 \leq i \leq t$,*

$$e(S_1, \ldots, S_t) = p \prod_{i=1}^{t} |S_i| + o\left(n^k\right)$$

where $e(S_1, \ldots, S_t)$ is the number of tuples $(s_1, \ldots, s_t)$ such that $s_1 \cup \cdots \cup s_t$ is a hyperedge and $s_i \in S_i$.

- **Count**[π-linear]: *If F is an f-vertex, m-edge, k-uniform, π-linear hypergraph, then the number of labeled copies of F in H_n is $p^m n^f + o(n^f)$.*

- **Cycle**$_4$[π]: *The number of labeled copies of $C_{\pi,4}$ in H_n is at most $p^{|E(C_{\pi,4})|} n^{|V(C_{\pi,4})|} + o(n^{|V(C_{\pi,4})|})$, where $C_{\pi,4}$ is the hypergraph four cycle of type π which is defined in Section 2.*

- **Cycle**$_{4\ell}$[π]: *the number of labeled copies of $C_{\pi,4\ell}$ in H_n is at most $p^{|E(C_{\pi,4\ell})|} n^{|V(C_{\pi,4\ell})|} + o(n^{|V(C_{\pi,4\ell})|})$, where $C_{\pi,4\ell}$ is the hypergraph cycle of type π and length 4ℓ defined in Section 2.*

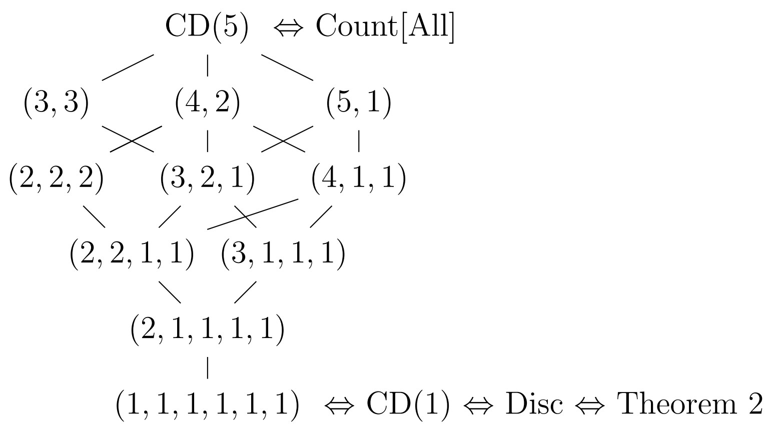


Figure 1: The Hasse diagram of quasirandom properties for $k = 6$.

Remarks.

- Theorem 3 is stated for hypergraph sequences $\mathcal{H} = \{H_n\}_{n\to\infty}$ where $|V(H_n)| = n$ for every n, but this requirement can be weakened. All the above theorems work perfectly well when considering sequences of hypergraphs whose number of vertices goes to infinity but perhaps might not be defined for every n. One way of enhancing the statements of these theorems to this situation is to transition away from the little-o notation and use an ϵ-δ type statement. Instead, following Chung, Graham, and Wilson [15], we maintain the language of sequences of hypergraphs as follows. Let $\mathcal{H} = \{H_{n_q}\}_{q\to\infty}$ be a sequence of hypergraphs such that $|V(H_{n_q})| = n_q$, $n_q < n_{q+1}$, and $|E(H_{n_q})| \geq p\binom{n_q}{k} + o(n_q^k)$, where now the little-$o$ expression means there exists a function $f(q)$ such that $|E(H_{n_q})| \geq p\binom{n_q}{k} + f(q)$ with $\lim_{q\to\infty} f(q) n_q^{-k} = 0$. Similarly, when we say that property P (which might include a little-o expression) implies a property P', what we mean is that there exist functions $f(q)$ and $f'(q)$ such that $P(f(q))$ implies $P'(f'(q))$, where the notation $P(f(q))$ stands for the property P with the little-o replaced by the function $f(q)$. In the rest of this paper, we ignore these complications and work directly with hypergraph sequences defined for every n but all the proofs extend into this world of hypergraph subsequences.

- If $\pi = 1 + \cdots + 1$, the partition of k into k ones, then the equivalences $\texttt{Expand}[\pi] \Leftrightarrow \texttt{Count}[\pi\text{-linear}] \Leftrightarrow \texttt{Cycle}_4[\pi]$ of Theorem 3 constitute Theorem 2.

- If π' is a refinement of π, then clearly $\texttt{Count}[\pi\text{-linear}] \Rightarrow \texttt{Count}[\pi'\text{-linear}]$ and so if $\{H_n\}_{n\to\infty}$ is a sequence satisfying the properties in Theorem 3 for π, it satisfies the properties for π'. In a companion paper [37], we show the converse: if π' is not a refinement of π then $\texttt{Expand}[\pi] \not\Rightarrow \texttt{Expand}[\pi']$ so the property $\texttt{Expand}[\pi]$ is distinct for distinct π and arranged in a poset via partition refinement. This is shown in Figure 1. In addition, [37] determines the entire relationship between essentially all quasirandom properties studied in the literature, completing a project suggested by Chung [8, 9].

- For graphs, a family $\mathcal{F}$ of graphs is called *p-forcing* if $\forall F \in \mathcal{F}$, the number of labeled copies of F in G is $(1+o(1))p^{|E(F)|}n^{|V(F)|}$ implies that G is quasirandom in the sense of Theorem 1. Theorem 1 proves that $\{K_2, C_{2t}\}$ is p-forcing and Skokan and Thoma [45] proved that $\{K_2, K_{s,t}\}$ is p-forcing for $s, t \geq 2$. Skokan and Thoma [45] also posed the forcing conjecture: for every bipartite F which is not a forest, $\{K_2, F\}$ is p-forcing. This conjecture is still open. Dellamonica et al. [19] investigated what properties of G are forced by $\{K_2, T\}$ when T is a tree, and Conlon, Fox, and Sudakov [16] proved the forcing conjecture holds for a class of bipartite graphs. Say that a family $\mathcal{F}$ of k-uniform hypergraphs is (π, p)-forcing if $\forall F \in \mathcal{F}$, the number of labeled copies of F in H is $(1 + o(1))p^{|E(F)|}n^{|V(F)|}$ implies that H is π-quasirandom in the sense of Theorem 3. Theorem 3 proves that $\{K_k^{(k)}, C_{\pi,4\ell}\}$ is (π, p)-forcing. It would be interesting to determine which families are (π, p)-forcing.

The remainder of this paper is organized as follows. In Section 2, we define the hypergraph cycles $C_{\pi,4}$. Section 3 gives the formal definition of eigenvalues with respect to π. Theorem 3 is proved by showing a chain of implications in the order stated in the theorem; Section 4 proves $\texttt{Eig}[\pi] \Rightarrow \texttt{Expand}[\pi]$, Section 5 proves $\texttt{Expand}[\pi] \Rightarrow \texttt{Count}[\pi\text{-linear}]$, and Section 6 completes the proof of Theorem 3 by showing that $\texttt{Cycle}_{4\ell}[\pi] \Rightarrow \texttt{Eig}[\pi]$. Lastly, Appendix A contains some linear algebra facts required for the proofs. Throughout this paper, we use the notation $[n] = \{1, \ldots, n\}$.

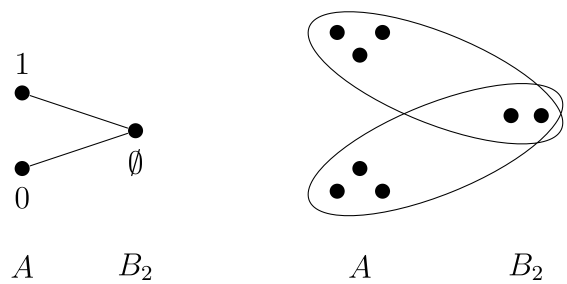
2 Hypergraph Cycles

In this section, we define the hypergraph cycles $C_{\pi,2\ell}$ used in Theorem 3. The hypergraph cycles $C_{\pi,2\ell}$ are defined by first defining steps, then defining a path as a combination of steps, and finally defining the cycle as a path with its endpoints identified.

Definition. *Let $\vec{\pi} = (1, \ldots, 1)$ be the ordered partition of t into t parts. Define the* step *of type $\vec{\pi}$, denoted $S_{\vec{\pi}}$, as follows. Let A be a vertex set of size 2^{t-1} where elements are labeled by binary strings of length $t-1$ and let $B_2, \ldots, B_t$ be disjoint sets of size 2^{t-2} where elements are labeled by binary strings of length $t-2$. The vertex set of $S_{\vec{\pi}}$ is the disjoint union $A \dot\cup B_2 \dot\cup \cdots \dot\cup B_t$. Make $\{a, b_2, \ldots, b_t\}$ a hyperedge of $S_{\vec{\pi}}$ if $a \in A$, $b_j \in B_j$, and the code for b_{j+1} is equal to the code formed by removing the jth bit of the code for a.*

For a general $\vec{\pi} = (k_1, \dots, k_t)$, start with $S_{(1,\dots,1)}$ and enlarge each vertex into the appropriate size; that is, a vertex in A is expanded into k_1 vertices and each vertex in B_j is expanded into k_j vertices. More precisely, the vertex set of $S_{\vec{\pi}}$ is $(A \times [k_1]) \dot\cup (B_2 \times [k_2]) \dot\cup \cdots \dot\cup (B_t \times [k_t])$, and if $\{a, b_2, \dots, b_t\}$ is an edge of $S_{(1,\dots,1)}$, then $\{(a,1), \dots, (a,k_1), (b_2,1), \dots, (b_2,k_2), \dots, (b_t,1), \dots, (b_t,k_t)\}$ is a hyperedge of $S_{\vec{\pi}}$.

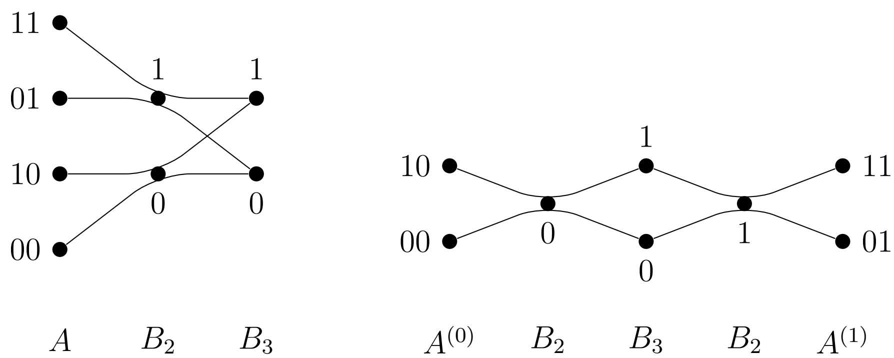
This defines the step of type $\vec{\pi}$, *denoted $S_{\vec{\pi}}$. Let $A^{(0)}$ be the ordered tuple of vertices of A in $S_{\vec{\pi}}$ whose binary code ends with zero and $A^{(1)}$ the ordered tuple of vertices of A whose binary code ends with one, where vertices are listed in lexicographic order within each $A^{(i)}$. These tuples $A^{(0)}$ and $A^{(1)}$ are the two* attach tuples of $S_{\vec{\pi}}$



(a) $S_{(1,1)}$ (b) $S_{(3,2)}$

Figure 2: Steps with $t = 2$

Figure 2 shows the steps of type $(1,1)$ and type $(3,2)$. Notice that each step has "length" two if we consider the attach tuples as the "ends" of a path.



(a) $S_{(1,1,1)}$ (b) $S_{(1,1,1)}$

Figure 3: Steps of type $\pi = (1,1,1)$

Figure 3 shows two different drawings of the step of type $\vec{\pi} = (1,1,1)$. Notice that the attach tuples are easily visible in Figure 3 *(b)*, since the two attach tuples are the codes in A ending with a zero and a one. The step of type $\vec{\pi} = (k_1, k_2, k_3)$ is an enlarged version of Figure 3 similar to Figure 2 *(b)*.

In general for arbitrary $\vec{\pi}$, the step $S_{\vec{\pi}}$ can be drawn in two ways similar to Figure 3. First from the definition, a step is a k-partite hypergraph with parts $A, B_2, \ldots, B_t$ so it can be drawn similar to Figure 3 *(a)*. But the step can also be drawn with the two attach tuples on separate ends of the picture like Figure 3 *(b)*. Let M_0 be the set of edges incident to vertices in the attach tuple $A^{(0)}$ and M_1 the set of edges incident to vertices in $A^{(1)}$. Edges from M_0 and M_1 intersect only in vertices in B_t because if $a_0 \in A^{(0)}$ and $a_1 \in A^{(1)}$ then the code for a_0 ends in a zero and the code for a_1 ends in a one, so only when deleting the last bit will the codes possibly be the same. Therefore, the step $S_{\vec{\pi}}$ can be viewed as a type of length two path in a hypergraph formed from a collection of k-partite edges M_0 between $A^{(0)}$ and B_t and another collection of k-partite edges M_1 between B_t and $A^{(1)}$. It is easy to see that all vertices in $\cup B_i$ have degree two and that the vertices in the attach tuples have degree one so are ready to be connected into longer paths.

Definition. *Let $\ell \geq 1$. The* path of type $\vec{\pi}$ of length 2ℓ, *denoted $P_{\vec{\pi},2\ell}$, is the hypergraph formed from ℓ copies of $S_{\vec{\pi}}$ with successive attach tuples identified. That is, let $T_1, \ldots, T_\ell$ be copies of $S_{\vec{\pi}}$ and let $A_i^{(0)}$ and $A_i^{(1)}$ be the attach tuples of T_i. The hypergraph $P_{\vec{\pi},2\ell}$ is the hypergraph consisting of $T_1, \ldots, T_\ell$ where the vertices of $A_i^{(1)}$ are identified with $A_{i+1}^{(0)}$ for every $1 \leq i \leq \ell - 1$. (Recall that by definition, $A_i^{(1)}$ and $A_{i+1}^{(0)}$ are tuples (i.e. ordered lists) of vertices, so the identification of $A_i^{(1)}$ and $A_{i+1}^{(0)}$ identifies the corresponding vertices in these tuples.) The* attach tuples of $P_{\vec{\pi},2\ell}$ *are the tuples $A_1^{(0)}$ and $A_\ell^{(1)}$.*

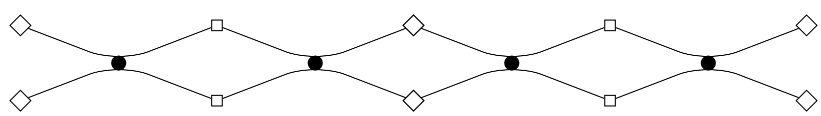


Figure 4: $P_{(1,1,1),4}$

In Figure 4, the path $P_{(1,1,1),4}$ is drawn as two copies of $S_{(1,1,1)}$ with attach tuples identified. The diamond, circle, and square vertices keep track of the parts A, B_2, B_3. For a general $P_{(k_1,k_2,k_3),4}$, each diamond vertex is enlarged into k_1 vertices, each circle vertex is enlarged into k_2 vertices, and each square vertex is enlarged into k_3 vertices. For a general $\vec{\pi}$, every step can be visualised as in Figure 3 *(b)* as two collections of k-partite edges M_0 and M_1 between A and B_t, so all paths $P_{\vec{\pi},2\ell}$ can be visualised as in Figure 4 as a concatenation of steps. Vertices in A have degree one and all other vertices have degree two, so the path has internal degree two vertices with degree one attach tuples.

Definition. *Let $\ell \geq 2$. The* cycle of type $\vec{\pi}$ and length 2ℓ, *denoted $C_{\vec{\pi},2\ell}$, is the hypergraph formed from $P_{\vec{\pi},2\ell}$ by identifying the attach tuples of $P_{\vec{\pi},2\ell}$.*

Lemma 4. *If π is an (unordered) proper partition of k and $\vec{\pi}$ and $\vec{\pi}'$ are two orderings of π, then $C_{\vec{\pi},2\ell} \cong C_{\vec{\pi}',2\ell}$.*

Proof. Let $\vec{\pi} = (k_1, \ldots, k_t)$ and let $\eta : \{2, \ldots, t\} \to \{2, \ldots, t\}$ be any bijection of the numbers $2, \ldots, t$. We first claim that $S_{\vec{\pi}} \cong S_{(k_1, k_{\eta(2)}, \ldots, k_{\eta(t)})}$. This follows directly from the definition of the step; the bit strings can be permuted using η. That is, the isomorphism between $S_{\vec{\pi}}$ and $S_{(k_1, k_{\eta(2)}, \ldots, k_{\eta(t)})}$ is the isomorphism which takes a vertex in A in $V(S_{\vec{\pi}})$ with binary code $a_2 \ldots a_t$ to the vertex with code $a_{\eta^{-1}(2)} \ldots a_{\eta^{-1}(t)}$ in A in $S_{(k_1, k_{\eta(2)}, \ldots, k_{\eta(t)})}$ and which also takes a vertex in B_j in $S_{\vec{\pi}}$ with code $b_2 \ldots b_{j-1} b_{j+1} \ldots b_t$ to the vertex with code $b_{\eta^{-1}(2)} b_{\eta^{-1}(3)} \ldots b_{\eta^{-1}(\eta(j)-1)} b_{\eta^{-1}(\eta(j)+1)} \ldots b_{\eta^{-1}(t)}$ in $B_{\eta(j)}$ in $S_{(k_1, k_{\eta(2)}, \ldots, k_{\eta(t)})}$. To see that this bijection preserves hyperedges, let $\{\vec{a}, \vec{b}_2, \ldots, \vec{b}_t\}$ be a set of vertices in $S_{\vec{\pi}}$ where $\vec{a} \in A$ and $\vec{b}_j \in B_j$. For every j, we have $a_2 \ldots a_{j-1} a_{j+1} \ldots a_t = b_2 \ldots b_{j-1} b_{j+1} \ldots b_t$ if and only if $a_{\eta^{-1}(2)} a_{\eta^{-1}(3)} \ldots a_{\eta^{-1}(\eta(j)-1)} a_{\eta^{-1}(\eta(j)+1)} \ldots a_{\eta^{-1}(t)} = b_{\eta^{-1}(2)} b_{\eta^{-1}(3)} \ldots b_{\eta^{-1}(\eta(j)-1)} b_{\eta^{-1}(\eta(j)+1)} \ldots b_{\eta^{-1}(t)}$. This implies that $\{\vec{a}, \vec{b}_2, \ldots, \vec{b}_t\}$ is an edge of $S_{\vec{\pi}}$ if and only if the image is an edge of $S_{(k_1, k_{\eta(2)}, \ldots, k_{\eta(t)})}$.

By the previous paragraph, it suffices to prove that $C_{\vec{\pi}, 2\ell} \cong C_{(k_t, k_2, \ldots, k_{t-1}, k_1), 2\ell}$ to complete the proof of the lemma. Indeed, if $\vec{\pi}' = (k_{f(1)}, \ldots, k_{f(t)})$ with $f(1) > 1$, then the transformations $(1, \ldots, t) \to (1, 2, \ldots, f(1)-1, f(1)+1, \ldots, t, f(1)) \to (f(1), 2, \ldots, f(1)-1, f(1)+1, \ldots, t, 1) \to (f(1), \ldots, f(t))$ show that $C_{\vec{\pi}, 2\ell} \cong C_{\vec{\pi}', 2\ell}$.

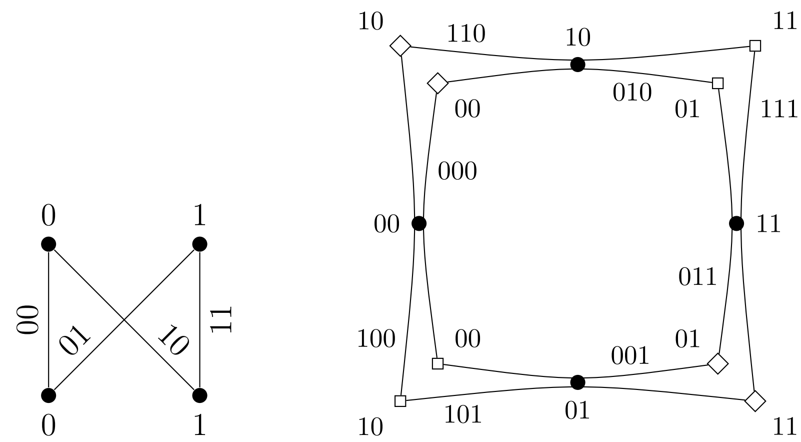
The fact that $C_{\vec{\pi}, 2\ell} \cong C_{(k_t, k_2, \ldots, k_{t-1}, k_1), 2\ell}$ is easy to see in Figure 4. In Figure 4, consider swapping the diamond and square vertices. This changes the path, but the cycle is Figure 4 with the diamond vertices on the ends identified, so swapping the diamond and square vertices preserves the cycle. In general, as discussed before, the step $S_{\vec{\pi}}$ can be drawn similar to Figure 3 *(b)* as a collection M_0 of k-partite edges between $A^{(0)}$ and B_t and a collection M_1 of k-partite edges between $A^{(1)}$ and B_t and so the path $P_{\vec{\pi}, 2\ell}$ can be visualized like Figure 4. Therefore the cycle $C_{\vec{\pi}, 2\ell}$ consists of a list of collections of k-partite edges; $E(C_{\vec{\pi}, 2\ell})$ is $M_{1,0} \dot\cup M_{1,1} \dot\cup M_{2,0} \dot\cup M_{2,1} \dot\cup \cdots \dot\cup M_{\ell,0} \dot\cup M_{\ell,1}$ where $M_{i,0} \cup M_{i,1}$ is a copy of $S_{\vec{\pi}}$. But $M_{i,1} \cup M_{i+1,0}$ (modulo ℓ) forms a copy of $S_{(k_t, k_2, \ldots, k_{t-1}, k_1)}$ for all i as follows. Since bit strings in the A-part of $M_{i,1}$ have last bit one by definition, drop the last bit. After dropping these bits, for each edge E in $M_{i,1}$, the vertices in $E \cap A$ and $E \cap B_t$ have the same code. Also, for $2 \le j \le t-1$ the code for the vertices $E \cap B_j$ is formed by adding a one to the bit string for $E \cap A$ and then deleting the $(j-1)$th entry. But since $E \cap A$ and $E \cap B_t$ have the same code, this is the same as adding a one to the bit string for $E \cap B_t$ and then deleting the $(j-1)$th entry. Therefore we could add a one to the end of all the codes in B_t and now have half of the edges which make up $S_{(k_t, k_2, \ldots, k_{t-1}, k_1)}$. A similar argument shows that $M_{i+1,0}$ forms the other half and so $M_{i,1} \cup M_{i+1,0}$ is a copy of $S_{(k_t, k_2, \ldots, k_{t-1}, k_1)}$. Thus $C_{\vec{\pi}, 2\ell}$ is built out of ℓ copies of $S_{(k_t, k_2, \ldots, k_{t-1}, k_1)}$. $\square$

Definition. *Let π be an (unordered) proper partition of k. The* cycle of type π and length 2ℓ*, denoted $C_{\pi, 2\ell}$, is $C_{\vec{\pi}, 2\ell}$ where $\vec{\pi}$ is any ordering of π.*

Definition. *Let $\ell \ge 2$. A* walk of type $\vec{\pi}$ and length 2ℓ *in a hypergraph H is a function $f : V(P_{\vec{\pi}, 2\ell}) \to V(H)$ that preserves edges. Informally, a walk is a path where the vertices are not necessarily distinct. A* circuit of type π of length 2ℓ *in a hypergraph H is a function $f : V(C_{\pi, 2\ell}) \to V(H)$ that preserves edges. Informally, a circuit is a cycle where the vertices are not necessarily distinct.*

There are two alternative definitions of the cycle of length four. First, Conlon et al. [17] defined a cycle of length four for $\pi = 1 + \cdots + 1$ by an operation called *reflection*. Our definition of $C_{1+\cdots+1,4}$ is equivalent to the definition in [17]; this can be seen by noticing that the bit strings in our definition keep track of the vertex duplications which occur during reflection.

Finally, there is a concise direct definition of the cycle of type π and length four which avoids the complexity of defining steps and paths. We will not use this shorter definition in this paper, instead working with steps, paths, and walks, but we include this short definition for completeness. Let $D_1, \ldots, D_t$ be disjoint sets of size 2^{t-1} whose elements are labeled by $(t-1)$-length binary strings. The vertex set is $D_1 \dot\cup \ldots \dot\cup D_t$. For $d_1 \in D_1, \ldots, d_t \in D_t$, make $\{d_1, \ldots, d_t\}$ a hyperedge if there exists a binary string s of length t such that the code for d_i equals the code formed by deleting the ith bit of s. The cycle for general π is formed by enlarging this cycle appropriately. Figure 5 shows cycles drawn using this definition.



(a) $C_{(1,1),4}$

(b) $C_{(1,1,1),4}$

Figure 5: Alternate definition of the cycle of length four

3 Hypergraph Eigenvalues

This section contains the definition of the largest and second largest eigenvalues of a hypergraph with respect to π and also contains some discussion and basic facts about them.

There have been three independently developed approaches to hypergraph eigenvalues: a definition by Chung [11] and Lu and Peng [39, 40] using matrices, an approach of Friedman and Wigderson [23, 24] and Cooper and Dutle [18], and lastly the eigenvalues of the shadow graph [7, 21, 41, 42, 47]. The definitions of Friedman and Wigderson [23, 24] are most suitable for our purposes and we will use their definitions as our starting point.

Definition. *Let $V_1, \ldots, V_k$ be finite-dimensional vector spaces over $\mathbb{R}$. A* k-linear map *is a function $\phi : V_1 \times \cdots \times V_k \to \mathbb{R}$ such that for each $1 \le i \le k$, ϕ is linear in the ith coordinate.*

That is, for every fixed $x_i \in V_i$, $\phi(x_1, \ldots, x_{i-1}, \cdot, x_{i+1}, \ldots, x_n)$ is a linear map from V_i to $\mathbb{R}$. A k-linear map $\phi : V^k \to \mathbb{R}$ is symmetric *if for all permutations η of $[k]$ and all $x_1, \ldots, x_k \in V$, $\phi(x_1, \ldots, x_k) = \phi(x_{\eta(1)}, \ldots, x_{\eta(k)})$.*

Definition. *Let $V_1, \ldots, V_k$ be finite-dimensional vector spaces over $\mathbb{R}$, let $B_i = \{b_{i,1}, \ldots, b_{i,\dim(V_i)}\}$ be an orthonormal basis of V_i, and let $\phi : B_1 \times \cdots \times B_k \to \mathbb{R}$ be any map.* Extending ϕ linearly to $V_1 \times \cdots \times V_k$ *means that ϕ is extended to a map $V_1 \times \cdots \times V_k \to \mathbb{R}$ where for $x_1 \in V_1, \ldots, x_k \in V_k$,*

$$\phi(x_1, \ldots, x_k) = \sum_{j_1=1}^{\dim(V_1)} \cdots \sum_{j_k=1}^{\dim(V_k)} \langle x_1, b_{1,j_1} \rangle \cdots \langle x_k, b_{k,j_k} \rangle \, \phi(b_{1,j_1}, \ldots, b_{k,j_k}). \tag{1}$$

Note that extending ϕ in this way produces a k-linear map.

Definition. ***(Friedman and Wigderson [23, 24])*** *Let H be a k-uniform hypergraph. The* adjacency map *of H is the symmetric k-linear map $\tau_H : W^k \to \mathbb{R}$ defined as follows, where W is the vector space over $\mathbb{R}$ of dimension $|V(H)|$. First, for all $v_1, \ldots, v_k \in V(H)$, let*

$$\tau_H(e_{v_1}, \ldots, e_{v_k}) = \begin{cases} 1 & \{v_1, \ldots, v_k\} \in E(H), \\ 0 & \textit{otherwise}, \end{cases}$$

where e_v denotes the indicator vector of the vertex v, that is the vector which has a one in coordinate v and zero in all other coordinates. We have defined the value of τ_H when the inputs are standard basis vectors of W. Extend τ_H to all the domain linearly.

Definition. *Let $W_1, \ldots, W_k$ be finite dimensional vector spaces over $\mathbb{R}$ and let $\phi : W_1 \times \cdots \times W_k \to \mathbb{R}$ be a k-linear map. The* spectral norm of ϕ *is*

$$\|\phi\| = \sup_{\substack{x_i \in W_i \\ \|x_i\|=1}} |\phi(x_1, \ldots, x_k)| \, .$$

Before defining the first and second largest eigenvalue of H with respect to a general partition π, we give the definitions when $\pi = 1 + \cdots + 1$, that is π is the partition into k ones. For codegree regular hypergraphs with loops (see the definition of loops in the discussion that follows), these are exactly the definitions of Friedman and Wigderson [23, 24].

Definition. *Let H be an n-vertex, k-uniform hypergraph, let W be the vector space over $\mathbb{R}$ of dimension n, and let $J : W^k \to \mathbb{R}$ be the all-ones map. That is, if $e_{i_1}, \ldots, e_{i_k}$ are any standard basis vectors of W, then $J(e_{i_1}, \ldots, e_{i_k}) = 1$, and J is extended linearly to all of the domain as in* (1)*.*

The largest eigenvalue of H with respect to $\pi = 1 + \cdots + 1$ *is $\|\tau_H\|$ and the* second largest eigenvalue of H with respect to $\pi = 1 + \cdots + 1$ *is $\left\| \tau_H - \frac{k!|E(H)|}{n^k} J \right\|$.*

In order to extend this definition to general $\vec{\pi} = (k_1, \dots, k_t)$, it is convenient to use the language of tensor products.

Definition. *Let V and W be finite dimensional vector spaces over $\mathbb{R}$ of dimension n and m respectively. The* tensor product of V and W, *written $V \otimes W$, is the vector space over $\mathbb{R}$ of dimension nm. A typical tensor a in $V \otimes W$ has the form $a = \sum_{i=1}^{\dim(V)} \sum_{j=1}^{\dim(W)} \alpha_{i,j}(e_i \otimes e'_j)$, where $\alpha_{i,j} \in \mathbb{R}$ and $e_1, \dots e_{\dim(V)}$ is the standard basis of V and $e'_1, \dots, e'_{\dim(W)}$ is the standard basis of W. The* length *of a tensor is the length of the vector in the vector space $V \otimes W$. Thus the length of a is $\left(\sum_{i=1}^{\dim(V)} \sum_{j=1}^{\dim(W)} \alpha_{i,j}^2\right)^{1/2}$.*

We are now ready to define the map $\tau_{\vec{\pi}}$ and then the first and second largest eigenvalue of H with respect to π for a general π. In the definition, think of the tensor product $W^{\otimes k_i}$ as a vector space of dimension $|V(H)|^{k_i}$ indexed by ordered k_i-sets of vertices.

Definition. *Let W be a finite dimensional vector space over $\mathbb{R}$, let $\sigma : W^k \to \mathbb{R}$ be any k-linear function, and let $\vec{\pi}$ be a proper ordered partition of k, so $\vec{\pi} = (k_1, \dots, k_t)$ for some integers $k_1, \dots, k_t$ with $t \geq 2$. Now define a t-linear function $\sigma_{\vec{\pi}} : W^{\otimes k_1} \times \cdots \times W^{\otimes k_t} \to \mathbb{R}$ by first defining $\sigma_{\vec{\pi}}$ when the inputs are basis vectors of $W^{\otimes k_i}$ and then extending linearly. For each i, $B_i = \{b_{i,1} \otimes \cdots \otimes b_{i,k_i} : b_{i,j}$ is a standard basis vector of $W\}$ is a basis of $W^{\otimes k_i}$, so for each i, pick $b_{i,1} \otimes \cdots \otimes b_{i,k_i} \in B_i$ and define*

$$\sigma_{\vec{\pi}}\left(b_{1,1} \otimes \cdots \otimes b_{1,k_1}, \dots, b_{t,1} \otimes \cdots \otimes b_{t,k_t}\right) = \sigma(b_{1,1}, \dots, b_{1,k_1}, \dots, b_{t,1}, \dots, b_{t,k_t}).$$

Now extend $\sigma_{\vec{\pi}}$ linearly to all of the domain. $\sigma_{\vec{\pi}}$ will be t-linear since σ is k-linear.

Definition. *Let H be a k-uniform hypergraph and let $\tau = \tau_H$ be the (k-linear) adjacency map of H. Let π be any (unordered) partition of k and let $\vec{\pi}$ be any ordering of π. The* largest and second largest eigenvalues of H with respect to π, *denoted $\lambda_{1,\pi}(H)$ and $\lambda_{2,\pi}(H)$, are defined as*

$$\lambda_{1,\pi}(H) := \|\tau_{\vec{\pi}}\| \quad \textit{and} \quad \lambda_{2,\pi}(H) := \left\| \tau_{\vec{\pi}} - \frac{k!|E(H)|}{n^k} J_{\vec{\pi}} \right\|.$$

Both $\lambda_{1,\pi}(H)$ and $\lambda_{2,\pi}(H)$ are well defined since τ and J are symmetric maps.

Remarks.

- When $k = 2$ and $\pi = 1 + 1$, the spectral norm of τ_H of a graph H exactly corresponds to the largest eigenvalue of H in absolute value in the usual definition of eigenvalues of graphs. The symmetric bilinear map τ_H is the map $\tau_H(x, y) = x^T Ay$ where A is the adjacency matrix of H. From linear algebra, we know that the spectral norm of τ_H is the largest eigenvalue of A in absolute value.
- When $k = 2$ and $\pi = 1 + 1$, our definition of second largest eigenvalue of a d-regular graph H is equivalent to the usual definition of second largest eigenvalue in absolute

value of the adjacency matrix A. If G is a d-regular graph, then $2|E(H)|/n^2 = \frac{d}{n}$, so our definition of the second largest eigenvalue of G is the spectral norm of $\tau_G - \frac{d}{n}J$. This bilinear map corresponds to the matrix $A - \frac{d}{n}J$ where A is the adjacency matrix and J is now the all-ones matrix. The largest eigenvalue of $A - \frac{d}{n}J$ in absolute value is the second largest eigenvalue of A in absolute value, and this equals the spectral norm of the respective map.

- The previous paragraph extends to d-codegree regular hypergraphs with loops, which were studied by Friedman and Wigderson [23, 24]. A *k-uniform hypergraph with loops* is a pair of sets $(V(H), E(H))$, where $V(H)$ is any finite set and $E(H)$ is a set of k-multisets of elements from $V(H)$. Informally, multiple edges are not allowed, each edge has size exactly k, and a vertex can be repeated multiple times inside an edge. A k-uniform hypergraph H with loops is *d-codegree regular* if for every $(k-1)$-multiset S of $V(H)$, there are exactly d hyperedges containing S. For such a hypergraph, $k!|E(H)|/n^k = d/n$ so $\lambda_{2,1+\cdots+1}(H) = \left\|\tau_H - \frac{d}{n}J\right\|$. This is exactly the definition of Friedman and Wigderson [23, 24]. In addition, they proved that if H is d-codegree regular with loops, then $\lambda_{1,1+\cdots+1}(H) = dn^{(k-2)/2}$ and is achieved by the all-ones vectors scaled to unit length. This generalizes the well known fact that the largest eigenvalue of a d-regular graph is d with eigenvector the all-ones vector. Friedman and Wigderson [23, 24] also proved several facts about $\lambda_{2,1+\cdots+1}(H)$ including upper and lower bounds, an Expander Mixing Lemma which we generalize to all π in Theorem 5, and the value of $\lambda_{2,1+\cdots+1}$ on the random hypergraph.

- If π' is a refinement of π, then it is easy to see that by combining tensors $\lambda_{1,\pi'}(H) \le \lambda_{1,\pi}(H)$ and $\lambda_{2,\pi'}(H) \le \lambda_{2,\pi}(H)$ for every H. Thus refining the partition does not increase $\lambda_{1,\pi}$ or $\lambda_{2,\pi}$, which matches exactly with the arrangement of quasirandom properties in Figure 1: $\texttt{Eig}[\pi] \Rightarrow \texttt{Eig}[\pi']$.

4 $\texttt{Eig}[\pi] \Rightarrow \texttt{Expand}[\pi]$

In this section we prove a generalization of the graph Expander Mixing Lemma which relates spectral and expansion properties of graphs. The graph version was first discovered independently by Alon and Milman [2] and Tanner [49]. For background on graph expansion and eigenvalues, see [1, 5, 30]. The following theorem extends on the hypergraph Expander Mixing Lemma of Friedman and Widgerson [23, 24], which applied for $\pi = 1 + \cdots + 1$. The theorem is stated for ordered partitions $\vec{\pi}$, but trivially gives the same result for any ordering $\vec{\pi}$ of a partition π.

Theorem 5. *(Hypergraph Expander Mixing Lemma) Let H be an n-vertex, k-uniform hypergraph. Let $\vec{\pi} = (k_1, \dots, k_t)$ be a proper ordered partition of k and let $S_i \subseteq \binom{V(H)}{k_i}$ for $1 \le i \le t$. Then*

$$\left| e(S_1, \dots, S_t) - \frac{k!|E(H)|}{n^k} \prod_{i=1}^{t} |S_i| \right| \le \lambda_{2,\pi}(H) \sqrt{|S_1| \cdots |S_t|},$$

where $e(S_1, \dots, S_t)$ is the number of ordered tuples $(s_1, \dots, s_t)$ such that $s_1 \cup \cdots \cup s_t \in E(H)$ and $s_i \in S_i$.

Proof. Let $q = \frac{k!|E(H)|}{n^k}$, let τ_H be the adjacency map of H, and let $\sigma = \tau_H - qJ$. It is easy to see that by definition, $(\tau - qJ)_{\vec{\pi}} = \tau_{\vec{\pi}} - qJ_{\vec{\pi}}$, so $\lambda_{2,\pi}(H) = \|\sigma_{\vec{\pi}}\|$. Let $\chi_{S_i} \in W^{k_i \otimes}$ be the indicator tensor of S_i. If we let $V(H) = [n]$, then

$$\chi_{S_i} = \sum_{\substack{\{v_1, \dots, v_{k_i}\} \in S_i \\ v_1 \le \cdots \le v_{k_i}}} (e_{v_1} \otimes \cdots \otimes e_{v_{k_i}}).$$

By the linearity of $\sigma_{\vec{\pi}}$ and the definition of $J_{\vec{\pi}}$,

$$\sigma_{\vec{\pi}}(\chi_{S_1}, \dots, \chi_{S_t}) = \tau_{\vec{\pi}}(\chi_{S_1}, \dots, \chi_{S_t}) - qJ_{\vec{\pi}}(\chi_{S_1}, \dots, \chi_{S_t}) = e(S_1, \dots, S_t) - q\prod_{i=1}^{t} |S_i|.$$

Before upper bounding this by $\lambda_{2,\pi}(H)$, we must scale each indicator tensor to be unit length. Since $\{e_{j_1} \otimes \cdots \otimes e_{j_{k_i}} : 1 \le j_1, \dots, j_{k_i} \le n\}$ forms a basis of $W^{\otimes k_i}$, we have $\|\chi_{S_i}\| = \sqrt{|S_i|}$. Thus

$$\left|\sigma_{\vec{\pi}}\left(\frac{\chi_{S_1}}{\|\chi_{S_1}\|}, \dots, \frac{\chi_{S_t}}{\|\chi_{S_t}\|}\right)\right| \le \|\sigma_{\vec{\pi}}\| = \lambda_{2,\pi}(H).$$

Consequently,

$$|\sigma_{\vec{\pi}}(\chi_{S_1}, \dots, \chi_{S_t})| \le \lambda_{2,\pi}(H)\, \|\chi_{S_1}\| \cdots \|\chi_{S_t}\| = \lambda_{2,\pi}(H)\sqrt{|S_1| \cdots |S_t|},$$

and the proof is complete. $\square$

Note that we could get an even tighter bound in Theorem 5 by projecting each χ_{S_i} onto the space perpendicular to the all-ones vector to keep the value of $\sigma_{\vec{\pi}}$ the same but shorten the length of the vectors given as input. Friedman and Widgerson [23, 24] compute this tighter bound for $\pi = 1 + \cdots + 1$; we leave this as an exercise to the reader for general π because the tighter bound is not required to prove `Eig`$[\pi] \Rightarrow$ `Expand`$[\pi]$.

Lemma 6. *Let $\mathcal{H} = \{H_n\}$ be a sequence of k-uniform hypergraphs with $|V(H_n)| = n$ and $|E(H_n)| \ge p\binom{n}{k} + o(n^k)$. Let τ_n be the adjacency map of H_n and let $\vec{\pi} = (k_1, \dots, k_t)$ be a proper ordered partition of k. If $\lambda_{1,\pi}(H_n) = pn^{k/2} + o(n^{k/2})$, then $|E(H_n)| = p\binom{n}{k} + o(n^k)$.*

Proof. Throughout this proof the subscripts on n are dropped for simplicity. Let W be the vector space over $\mathbb{R}$ of dimension n. For $1 \le i \le t$, let $\vec{1}_{k_i}$ denote the all-ones vector in $W^{\otimes k_i}$, so $\|\vec{1}_{k_i}\| = n^{k_i/2}$. Then

$$\begin{aligned}
\tau_{\vec{\pi}}\left(\frac{\vec{1}_{k_1}}{n^{k_1/2}}, \dots, \frac{\vec{1}_{k_t}}{n^{k_t/2}}\right) &= \frac{1}{n^{k/2}}\tau_{\vec{\pi}}\left(\vec{1}_{k_1}, \dots, \vec{1}_{k_t}\right) = \frac{1}{n^{k/2}}\tau(\vec{1}_1, \dots, \vec{1}_1) \\
&= \frac{1}{n^{k/2}} \sum_{i_1, \dots, i_k = 1}^{n} \tau(e_{i_1}, \dots, e_{i_k}) \\
&= \frac{1}{n^{k/2}} k!|E(H)|.
\end{aligned}$$

Thus the spectral norm of $\tau_{\vec{\pi}}$ is at least $k!|E(H)|/n^{k/2}$, so

$$pn^{k/2} \leq \frac{k!|E(H)|}{n^{k/2}} + o(n^{k/2}) \leq \|\tau_{\vec{\pi}}\| + o(n^{k/2}) = pn^{k/2} + o(n^{k/2})$$

This implies equality (up to $o(n^{k/2})$) throughout the above expression. In particular, $|E(H_n)| = p\binom{n}{k} + o(n^k)$. $\square$

Proofthat $\texttt{Eig}[\pi] \Rightarrow \texttt{Expand}[\pi]$. First, $\texttt{Eig}[\pi]$ contains the assertion that $\lambda_{1,\pi}(H_n) = pn^{k/2} + o(n^{k/2})$ which by Lemma 6 implies $|E(H_n)| = p\binom{n}{k} + o(n^k)$. Consequently, $k!|E(H_n)|/n^k = (1 + o(1))p$ and Theorem 5 imply that

$$\left|e(S_1, \ldots, S_t) - (1 + o(1))p \prod_{i=1}^{t} |S_i|\right| \leq \lambda_{2,\pi}(H)\sqrt{|S_1| \cdots |S_t|} \tag{2}$$

for any choice of $S_i \subseteq \binom{V(H_n)}{k_i}$, $i = 1, \ldots, t$. Since π is a partition of k, $\sqrt{|S_1| \cdots |S_t|} = O(n^{k/2})$. Also, $\texttt{Eig}[\pi]$ states that $\lambda_{2,\pi}(H) = o(n^{k/2})$. Thus (2) becomes

$$\left|e(S_1, \ldots, S_t) - p|S_1| \cdots |S_t|\right| = o(n^k),$$

which proves $\texttt{Expand}[\pi]$. $\square$

5 $\texttt{Expand}[\pi] \Rightarrow \texttt{Count}[\pi\texttt{-linear}]$

In this section we prove $\texttt{Expand}[\pi] \Rightarrow \texttt{Count}[\pi\text{-linear}]$ by proving an embedding lemma for hypergraphs. The proof below is a generalization of an argument by Kohayakawa et al. [32] who proved it in the special case of linear hypergraphs. The lemma below is proved for ordered partitions $\vec{\pi}$, but it is easy to see that the lemma is independent of the ordering chosen for $\vec{\pi}$.

Proposition 7. *Let $\vec{\pi} = (k_1, \ldots, k_t)$ be a proper ordered partition of k, let $0 < p < 1$, and let F be any fixed k-uniform, π-linear hypergraph with f vertices and m edges.*

Let $\mathcal{H} = \{H_n\}_{n\to\infty}$ be a sequence of k-uniform hypergraphs with $|V(H_n)| = n$, $|E(H_n)| = p\binom{n}{k} + o(n^k)$, and for which $\texttt{Expand}[\vec{\pi}]$ *holds. In other words, for every $S_1 \subseteq \binom{V(H)}{k_1}, \ldots, S_t \subseteq \binom{V(H)}{k_t}$, we have $e(S_1, \ldots, S_t) = p|S_1| \cdots |S_t| + o(n^k)$. Then the number of labeled copies of F in H is $p^m n^f + o(n^f)$.*

Proof. The proof is by induction on the number of edges of F. If F has zero or one edge, then the result is trivial. So assume that F has at least two edges and let E be the last edge in the ordering provided by the π-linearity of F. Let F_* be the hypergraph formed by deleting all vertices of E from F. Let Q_* be a labeled copy of F_* in H by which we mean that Q_* is an injective edge preserving map $Q_* : V(F_*) \to V(H)$. We can count the number of labeled copies of F in H by counting for each Q_*, the number of ways Q_* extends to a labeled copy of F. More precisely, we count the number of edge preserving injections $Q : V(F) \to V(H)$ which when restricted to $V(F_*)$ match the injection Q_*. Since F is π-linear, the edge E can be divided into $A_1, \ldots, A_t$ such that $|A_i| = k_i$ and the edges of F intersecting E can be divided into sets $R_1, \ldots, R_t$ such that every edge in R_i intersects E in a subset of A_i.

Consider some Q_* in H; we will count how many ways it extends to a labeled copy of F. For $1 \leq i \leq t$, define $S_i(Q_*)$ to be the following collection of k_i-sets. Let $Y \subseteq V(H)$ be a set of k_i vertices and add Y to $S_i(Q_*)$ if $Y \cap Im(Q_*) = \emptyset$ and there exists an edge preserving injection $V(F_*) \cup A_i \to Im(Q_*) \cup Y$ which when restricted to $V(F_*)$ matches the map Q_*. More informally, $S_i(Q_*)$ consists of all k_i-sets Y of vertices which can be used to extend Q_* to embed a labeled copy of $F_* \cup R_i$.

Every edge counted by $e(S_1(Q_*), \ldots, S_t(Q_*))$ creates several labeled copies of F which extend Q_*. First, let Δ_i be the number of edge preserving bijections $V(F_*) \cup A_i \to V(F_*) \cup A_i$ which are the identity map when restricted to $V(F_*)$. More informally, if we are given a non-labeled $F_* \cup R_i$ together with a labeling of the vertices of F_*, Δ_i is the number of ways of labeling the vertices of A_i. The numbers $\Delta_1, \ldots, \Delta_t$ are fixed numbers depending only on F; Δ_i depends on the way that edges in R_i intersect. For example, if every edge in R_i meets E in exactly A_i, then $\Delta_i = k_i!$. If the edges in R_i intersect differently, Δ_i will change but still depend only on F. Now we count the number of edge preserving injections $Q : V(F) \to V(H)$ where $Q|_{V(F_*)} = Q_*$ as follows. First pick an edge to use for E; this consists of picking one of the edges which take a k_1-set Y_1 from $S_1(Q_*)$, a k_2-set Y_2 from $S_2(Q_*)$, and so on. There are exactly $e(S_1(Q_*), \ldots, S_t(Q_*))$ such edges. We are embedding a labeled copy of F so next we order the vertices inside the set Y_1 chosen from $S_1(Q_*)$; there are Δ_1 ways of ordering the vertices of Y_1. Similarly, we order the vertices inside the other sets chosen from $S_i(Q_*)$ for a total of $\prod \Delta_i$ orderings. These are all the labeled copies of F which extend Q_*, so there are exactly $e(S_1(Q_*), \ldots, S_t(Q_*)) \prod \Delta_i$ labeled copes of F extending Q_*, i.e. exactly $e(S_1(Q_*), \ldots, S_t(Q_*)) \prod \Delta_i$ edge preserving injections $V(F) \to V(H)$ which when restricted to $V(F_*)$ are Q_*. By assupmtion,

$$e\big(S_1(Q_*), \ldots, S_t(Q_*)\big) = p|S_1(Q_*)| \cdots |S_t(Q_*)| + o(n^k).$$

Therefore, we can count the number of labeled copies of F in H by summing the value of $e(S_1(Q_*), \ldots, S_t(Q_*)) \prod \Delta_i$ over all Q_* in H. By the notation $\#\{F \,\mathrm{in}\, H\}$ we mean the number of labeled copies of F in H.

$$\begin{aligned}
\#\{F \,\mathrm{in}\, H\} &= \sum_{Q_*} e(S_1(Q_*), \ldots, S_t(Q_*)) \prod_i \Delta_i \\
&= \sum_{Q_*} \left(p\,|S_1(Q_*)| \cdots |S_t(Q_*)| \prod_i \Delta_i + o(n^k) \right) \\
&= p \sum_{Q_*} |S_1(Q_*)| \cdots |S_t(Q_*)| \prod_i \Delta_i + o(n^f), \qquad (3)
\end{aligned}$$

since the number of Q_* is bounded above by $n^{|V(F_*)|} = n^{f-k}$ so summing the term $o(n^k)$ over Q_* is bounded by $o(n^f)$. Let F_- be the hypergraph formed by removing the edge E from F

but keeping the same vertex set. Then similarly to the above, the number of labeled copies of F_- extending Q_* is $|S_1(Q_*)|\cdots|S_t(Q_*)|\prod\Delta_i$ since every labeled copy of F_- is formed by picking a set Y_1 from $S_1(Q_*)$, ordering it in Δ_i ways, picking a set Y_2 from $S_2(Q_*)$ and ordering it, and so on. Therefore the number of labeled copies of F_- in H is counted by summing over Q_* and counting $|S_1(Q_*)|\cdots|S_t(Q_*)|\prod\Delta_i$. Thus (3) continues as

$$\#\{F\,\mathrm{in}\,H\} = p\,\#\{F_-\ \mathrm{in}\,H\} + o(n^f). \tag{4}$$

By hypothisis, F is π-linear so F_- is π-linear. Thus by induction the number of labeled copies of F_- in H is $p^{m-1}n^f + o(n^f)$. Inserting this into (4) shows that the number of labeled copies of F in H is $p^m n^f + o(n^f)$. □

6 $\texttt{Cycle}_{4\ell}[\pi] \Rightarrow \texttt{Eig}[\pi]$

In Section 6.1, we state two crucial propositions on products and powers of multilinear maps and then show that these two propositions imply that $\texttt{Cycle}_{4\ell}[\pi] \Rightarrow \texttt{Eig}[\pi]$. The remainder of Section 6 is concerned with the proof of one of these propositions (the other is proved in the appendix). In Section 6.2 we discuss the special case of $\vec{\pi} = (3,2,1)$, in Section 6.3 we state the definitions of products and powers of multilinear maps, and finally in Section 6.4 we show that the powers of the adjacency map counts walks in a hypergraph.

6.1 Overview

First, let us recall the proof of $\texttt{Cycle}_4[1,1] \Rightarrow \texttt{Eig}[1,1]$ for graphs. If A is the adjacency matrix of a graph G, then A^4 counts walks of length 4 in the sense that the (i,j)-th entry of A^4 is the number of walks of length 4 between i and j. The trace of A^4 is then the number of circuits of length 4 in G. The trace of a square real symmetric matrix is the sum of its eigenvalues. If G is d-regular, then the largest eigenvalue of A^4 is d^4 so if the number of circuits of length four in G is $d^4 + o(n^4)$, then the trace of A^4 is $d^4 + o(n^4)$ which implies that all the eigenvalues of A besides d are $o(n)$. The proof for non-regular graphs is a straightforward generalization of this. Our proof for hypergraphs follows the same outline once some algebraic facts about multilinear maps are proved.

The first step in the general proof of $\texttt{Cycle}_{4\ell}[\pi] \Rightarrow \texttt{Eig}[\pi]$ is to define the product of two multilinear maps. This is a generalization of matrix multiplication. Next, we define the power of a multilinear map and show that if τ is the adjacency map of a hypergraph, then powers of $\tau_{\vec{\pi}}$ count walks. Since circuits are walks where the end points are identified, counting circuits will correspond to the trace of a power of $\tau_{\vec{\pi}}$. In Section 6.3, we define for any t-linear map ψ a square symmetric matrix $A[\psi^{2^{t-1}}]$. We write $A[\psi^{2^{t-1}}]^\ell$ for the ℓ-th matrix power of $A[\psi^{2^{t-1}}]$. The key technical proposition is the following.

Proposition 8. *Let H be a k-uniform hypergraph, let $\vec{\pi}$ be a proper ordered partition of k, and let $\ell \geq 2$ be an integer. Let τ be the adjacency map of H. Then $Tr\left[A[\tau_{\vec{\pi}}^{2^{t-1}}]^\ell\right]$ is the number of labeled circuits of type $\vec{\pi}$ and length 2ℓ in H.*

In Section 6.2, we sketch a proof of this proposition for the special case of $\vec{\pi} = (3, 2, 1)$ and in Section 6.4 give the proof for all $\vec{\pi}$. Next we need an algebraic property of the eigenvalues of multilinear maps.

Proposition 9. *Let $\{\psi_r\}_{r\to\infty}$ be a sequence of symmetric k-linear maps, where $\psi_r : V_r^k \to \mathbb{R}$, V_r is a vector space over $\mathbb{R}$ of finite dimension, and $\dim(V_r) \to \infty$ as $r \to \infty$. Let $\hat{1}$ denote the all-ones vector in V_r scaled to unit length and let $J : V_r^k \to \mathbb{R}$ be the k-linear all-ones map. Let π be a proper (unordered) partition of k, and assume that for every ordering $\vec{\pi}$ of π,*

$$\begin{aligned} \lambda_1(A[\psi_{\vec{\pi}}^{2^{t-1}}]) &= (1 + o(1))\psi\left(\hat{1}, \dots, \hat{1}\right)^{2^{t-1}}, \\ \lambda_2(A[\psi_{\vec{\pi}}^{2^{t-1}}]) &= o\left(\lambda_1(A[\psi_{\vec{\pi}}^{2^{t-1}}])\right). \end{aligned}$$

Then for every ordering $\vec{\pi}$ of π,

$$\|\psi_{\vec{\pi}} - qJ_{\vec{\pi}}\| = o(\psi(\hat{1}, \dots, \hat{1})),$$

where $q = \dim(V_r)^{-k/2}\psi(\hat{1}, \dots, \hat{1})$.

For graphs, $A[\tau^2]$ is the adjacency matrix squared so Proposition 9 states that the spectral norm of $A - \frac{2|E(G)|}{n^2}J$ is little-o of the square root of the largest eigenvalue of A^2, a fact proved by Chung, Graham, and Wilson (see the bottom of page 350 in [15]). Note that in graphs, if the graph sequence is assumed to be d-regular then Proposition 9 becomes trivial since the spectral norm of $A - \frac{d}{n}J$ equals the second largest eigenvalue of A. Similarly, if Theorem 3 is restricted to hypergraph sequences which are codegree regular, the proof of Proposition 9 becomes much simpler. Finally, we need one last simple property of multilinear maps.

Lemma 10. *Let $V_1, \dots, V_t$ be vector spaces over $\mathbb{R}$ and let $\phi : V_1 \times \cdots \times V_t \to \mathbb{R}$ be a t-linear map. Then $\|\phi\|^{2^{t-1}} \leq \lambda_1(A[\phi^{2^{t-1}}])$.*

Propositions 8 and 9 and Lemma 10 combine to prove that $\texttt{Cycle}_{4\ell}[\pi] \Rightarrow \texttt{Eig}[\pi]$ for any proper partition π as follows. The proof can be understood without knowing the definition of $A[\tau_{\vec{\pi}}^{2^{t-1}}]$ (which appears in Section 6.3): $A[\tau_{\vec{\pi}}^{2^{t-1}}]$ is a square symmetric matrix.

Proofthat $\texttt{Cycle}_{4\ell}[\pi] \Rightarrow \texttt{Eig}[\pi]$. Let $\mathcal{H} = \{H_n\}_{n\to\infty}$ be a sequence of hypergraphs and let τ_n be the adjacency map of H_n. For notational convenience, the subscript on n is dropped below. Throughout this proof, we use $\hat{1}$ to denote the all-ones vector scaled to unit length. Wherever we use the notation $\hat{1}$, it is the input to a multilinear map and so $\hat{1}$ denotes the all-ones vector in the appropriate vector space corresponding to whatever space the map is expecting as input. This means that in the equations below $\hat{1}$ can stand for different vectors in the same expression, but attempting to subscript $\hat{1}$ with the vector space (for example $\hat{1}_{V_3}$) would be notationally awkward.

The proof that $\texttt{Cycle}_{4\ell}[\pi] \Rightarrow \texttt{Eig}[\pi]$ comes down to checking the conditions of Proposition 9. Let $\vec{\pi}$ be any ordering of the entries of π. We will show that the first and second largest eigenvalues of $A = A[\tau_{\vec{\pi}}^{2^{t-1}}]$ are separated. Let $m = |E(C_{\pi,4\ell})| = 2\ell 2^{t-1}$ and note that $|V(C_{\pi,4\ell})| = mk/2$ since $C_{\pi,4\ell}$ is two-regular. A is a square symmetric real valued matrix, so let $\mu_1, \ldots, \mu_d$ be the eigenvalues of A arranged so that $|\mu_1| \geq \cdots \geq |\mu_d|$, where $d = \dim(A)$. The eigenvalues of $A^{2\ell}$ are $\mu_1^{2\ell}, \ldots, \mu_d^{2\ell}$ and the trace of $A^{2\ell}$ is $\sum_i \mu_i^{2\ell}$. Since all $\mu_i^{2\ell} \geq 0$, Proposition 8 and $\texttt{Cycle}_{4\ell}[\pi]$ implies that

$$\mu_1^{2\ell} + \mu_2^{2\ell} \leq \mathrm{Tr}\left[A^{2\ell}\right] = \#\{\text{possibly degenerate } C_{\pi,4\ell} \text{ in } H_n\} \leq p^m n^{mk/2} + o(n^{mk/2}). \tag{5}$$

We now verify the conditions on μ_1 and μ_2 in Proposition 9, and to do that we need to compute $\tau(\hat{1}, \ldots, \hat{1})$. The computations in Lemma 6 show that

$$\tau(\hat{1}, \ldots, \hat{1}) = \tau_{\vec{\pi}}(\hat{1}, \ldots, \hat{1}) = \frac{k!E(H)}{n^{k/2}}. \tag{6}$$

Using that $|E(H_n)| \geq p\binom{n}{k} + o(n^k)$, Lemma 10, and $\mu_1^{2\ell} \leq p^m n^{mk/2} + o(n^{mk/2})$ from (5),

$$pn^{k/2} + o(n^{k/2}) \leq \frac{k!E(H)}{n^{k/2}} = \tau_{\vec{\pi}}(\hat{1}, \ldots, \hat{1}) \leq \|\tau_{\vec{\pi}}\| \leq \mu_1^{1/2^{t-1}} \leq pn^{k/2} + o(n^{k/2}). \tag{7}$$

This implies equality up to $o(n^{k/2})$ throughout the above expression, so $\tau(\hat{1}, \ldots, \hat{1}) = pn^{k/2} + o(n^{k/2})$, $\lambda_{1,\pi}(H_n) = \|\tau_{\vec{\pi}}\| = pn^{k/2} + o(n^{k/2})$, and $\mu_1 = p^{2^{t-1}} n^{k2^{t-2}} + o(n^{k2^{t-2}})$, so $\mu_1 = (1 + o(1))\tau(\hat{1}, \ldots, \hat{1})^{2^{t-1}}$.

Insert $\mu_1 = p^{2^{t-1}} n^{k2^{t-2}} + o(n^{k2^{t-2}})$ into (5) to show that $\mu_2 = o(n^{k2^{t-2}})$. Therefore, the conditions of Proposition 9 are satisfied, so

$$\|\tau_{\vec{\pi}} - qJ_{\vec{\pi}}\| = o(\tau(\hat{1}, \ldots, \hat{1})) = o(n^{k/2}),$$

where $q = n^{-k/2}\tau(\hat{1}, \ldots, \hat{1})$. Using (6), $q = k!|E(H)|/n^k$. Thus $\|\tau_{\vec{\pi}} - qJ_{\vec{\pi}}\| = \lambda_{2,\pi}(H_n)$ and the proof is complete. $\square$

The above proof can be extended to even length cycles in the case when $\vec{\pi} = (k_1, k_2)$ is a partition into two parts. For these $\vec{\pi}$, the matrix $A[\tau_{\vec{\pi}}^2]$ can be shown to be positive semidefinite since $A[\tau_{\vec{\pi}}^2]$ will equal MM^T where M is the matrix associated to the bilinear map $\tau_{\vec{\pi}}$. Since $A[\tau_{\vec{\pi}}^2]$ is positive semidefinite, each $\mu_i \geq 0$ so any power of μ_i is non-negative. For partitions into more than two parts, we don't know if the matrix $A[\tau_{\vec{\pi}}^{2^{t-1}}]$ is always positive semidefinite or not.

6.2 Counting circuits in six uniform hypergraphs

In this section, we explain the key ideas behind Proposition 8 by looking at the case of six uniform hypergraphs with $\vec{\pi} = (3, 2, 1)$. Throughout this section, for a finite dimensional vector space W over $\mathbb{R}$, we denote by $e_1, \ldots, e_{\dim(W)}$ the standard basis vectors; e_i is a vector with a one in the ith coordinate and a zero everywhere else.

First, return to matrices and bilinear maps and let A and B be square matrices and let ϕ and ψ be the associated bilinear maps so $\phi(x, y) := x^T Ay$ and $\psi(x, y) := x^T By$. Now translate the matrix product AB into the language of bilinear maps: consider the i, jth coordinate of AB:

$$\phi\psi(e_i, e_j) = e_i^T A B e_j = \sum_{p=1}^{n} (e_i^T A e_p)(e_p^T B e_j) = \sum_{p=1}^{n} \phi(e_i, e_p)\psi(e_p, e_j) \tag{8}$$

since the i, j-th entry is the dot product of the ith row of A with the jth column of B and $e_i^T A e_p$ is the pth entry of the ith row of A and $e_p^T B e_j$ is the pth entry of the jth column of B. If both ϕ and ψ are the adjacency map, then (8) counts the number of walks of length two between i and j, since the product sums over all possible vertices p.

Now consider a six-uniform, n vertex hypergraph H and let $\tau = \tau_H$ be the adjacency map of H. Let W be a vector space over $\mathbb{R}$ of dimension $n = |V(H)|$. Consider the map $\tau_{\vec{\pi}}$ defined in Section 3:

$$\tau_{\vec{\pi}} : (W \otimes W \otimes W) \times (W \otimes W) \times W \to \mathbb{R}$$

$$\tau_{\vec{\pi}}(e_u \otimes e_v \otimes e_w, e_x \otimes e_y, e_z) = \tau(e_u, e_v, e_w, e_x, e_y, e_z) = \begin{cases} 1 & \text{if } \{u, v, w, x, y, z\} \in E(H) \\ 0 & \text{otherwise} \end{cases}$$

(extended linearly to all of the domain). Motivated by the form of matrix multiplication in equation (8), define the product of $\tau_{\vec{\pi}}$ with itself as the following map. This is a special case of the more general definition in Section 6.3. Let $a, a' \in W^{\otimes 3}$ so $a \otimes a' \in W^{\otimes 6}$ and let $b, b' \in W^{\otimes 2}$ so $b \otimes b' \in W^{\otimes 4}$. Define $\tau_{\vec{\pi}} * \tau_{\vec{\pi}}$ as follows:

$$\tau_{\vec{\pi}} * \tau_{\vec{\pi}} : W^{\otimes 6} \times W^{\otimes 4} \to \mathbb{R}$$

$$\tau_{\vec{\pi}} * \tau_{\vec{\pi}}(a \otimes a', b \otimes b') := \sum_{i=1}^{n} \tau_{\vec{\pi}}(a, b, e_i)\tau_{\vec{\pi}}(a', b', e_i), \tag{9}$$

where the definition is extended linearly to all of the domain. The product $\tau_{\vec{\pi}} * \tau_{\vec{\pi}}(a \otimes a', b \otimes b')$ will count the number of labeled copies of the hypergraph in Figure 6.

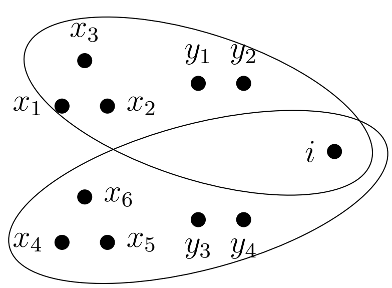


Figure 6: Hypergraph counted by $\tau_{\vec{\pi}} * \tau_{\vec{\pi}}$

Let $x_1, \ldots, x_6, y_1, \ldots, y_4$ be not necessarily distinct vertices and define $a = e_{x_1} \otimes e_{x_2} \otimes e_{x_3}$,

$a' = e_{x_4} \otimes e_{x_5} \otimes e_{x_6}$, $b = e_{y_1} \otimes e_{y_2}$, and $b' = e_{y_3} \otimes e_{y_4}$. Then

$$\tau_{\vec{\pi}} * \tau_{\vec{\pi}}(a \otimes a', b \otimes b') = \sum_{i=1}^{n} \tau(e_{x_1}, e_{x_2}, e_{x_3}, e_{y_1}, e_{y_2}, e_i)\tau(e_{x_4}, e_{x_5}, e_{x_6}, e_{y_3}, e_{y_4}, e_i).$$

Thus $\tau_{\vec{\pi}} * \tau_{\vec{\pi}}(a \otimes a', b \otimes b')$ counts the number of labeled copies of Figure 6 where vertex i is allowed to range over any vertex and the other vertices are fixed.

Next, we define the power of $\tau_{\vec{\pi}}^4$ which will count steps of type $\vec{\pi}$. The following definition of $\tau_{\vec{\pi}}^4$ is a special case of a more general definition in Section 6.3. Let $a, a', a'', a''' \in W^{\otimes 3}$ and define $\tau_{\vec{\pi}}^4$ as follows:

$$\tau_{\vec{\pi}}^4 : W^{\otimes 12} \to \mathbb{R}$$

$$\tau_{\vec{\pi}}^4(a \otimes a' \otimes a'' \otimes a''') := \sum_{i,j,\ell,m=1}^{n} \tau_{\vec{\pi}}^2(a \otimes a', e_i \otimes e_j \otimes e_\ell \otimes e_m)\tau_{\vec{\pi}}^2(a'' \otimes a''', e_i \otimes e_j \otimes e_\ell \otimes e_m).$$

Note that the definition still takes a similar form to (8) and (9) since the expression for $(\tau_{\vec{\pi}}^2)^2$ sums over a basis of $W^{\otimes 4}$ and takes the product of $\tau_{\vec{\pi}}^2$ with $\tau_{\vec{\pi}}^2$. Expand out the definition of $\tau_{\vec{\pi}}^2$ to obtain

$$\begin{aligned} &\sum_{i,j,\ell,m=1}^{n} \left(\sum_{s=1}^{n} \tau_{\vec{\pi}}(a, e_i \otimes e_j, e_s)\tau_{\vec{\pi}}(a', e_\ell \otimes e_m, e_s) \right) \tau_{\vec{\pi}}^2(a'' \otimes a''', e_i \otimes e_j \otimes e_\ell \otimes e_m) \\ &= \sum_{i,j,\ell,m,r,s=1}^{n} \tau_{\vec{\pi}}(a, e_i \otimes e_j, e_s)\tau_{\vec{\pi}}(a', e_\ell \otimes e_m, e_s)\tau_{\vec{\pi}}(a'', e_i \otimes e_j, e_r)\tau_{\vec{\pi}}(a''', e_\ell \otimes e_m, e_r). \qquad (10) \end{aligned}$$

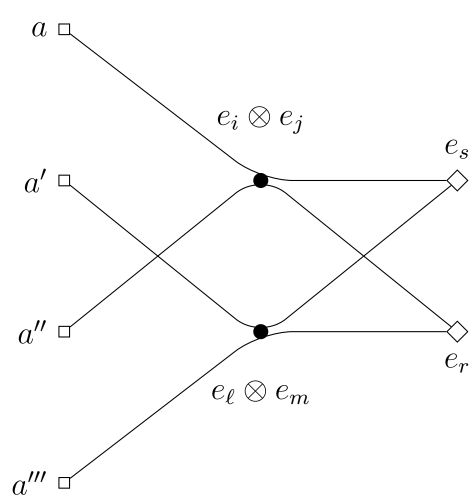


Figure 7: $S_{\vec{\pi}}$, the hypergraph counted by $\tau_{\vec{\pi}}^4$

Compare (10) and Figure 7. In the figure, each square vertex is actually a small cluster of three vertices, each filled circle is a small cluster of two vertices, and each diamond is a single vertex (recall that the step was defined by expanding Figure 3). Pick twelve vertices (four groups of three vertices) for the squares in Figure 7 and define $a, a', a'', a''' \in W^{\otimes 3}$ as indicator tensors for these vertex groups. That is, $a = e_{x_1} \otimes e_{x_2} \otimes e_{x_3}$ if x_1, x_2, x_3 are the three vertices specified for the topmost square in Figure 7. Once we specify the twelve vertices for the squares and define a, a', a'', a''', then $\tau^4_{\vec{\pi}}(a \otimes a' \otimes a'' \otimes a''')$ counts the number of copies of the step of type $\vec{\pi}$ with the square vertices fixed and the circle and diamond vertices allowed to range over all of $V(H)$. This is easy to see from (10), since the expression sums over i, j, ℓ, m, r, s and has a one if and only if all four edges drawn in the figure appear. Also, Figure 7 is drawn slightly differently but is the same as the hypergraph in Figure 3 so is the step $S_{\vec{\pi}}$. Lastly, notice that Figure 7 is two copies of Figure 6: one copy using a, a', and s and another using a'', a''', and r. This reflects the fact that $\tau^4_{\vec{\pi}}$ sums over the product of two $\tau^2_{\vec{\pi}}$.

Next, we would like to rearrange the inputs to $\tau^4_{\vec{\pi}}$ to separate the attach tuples of the step. Note that $a \otimes a''$ is one of the attach tuples and $a' \otimes a'''$ is the other attach tuple (each attach tuple is a tuple of six vertices). We would like to define a map which is the same as the map $\tau^4_{\vec{\pi}}$, but takes the 12-vertex tuple as two 6-vertex tuples of attach tuples. Define $A[\tau^4_{\vec{\pi}}]$ to be the following matrix/bilinear map.

$$A[\tau^4_{\vec{\pi}}] : W^{\otimes 6} \times W^{\otimes 6} \to \mathbb{R}$$
$$A[\tau^4_{\vec{\pi}}](a \otimes a'', a' \otimes a''') := \tau^4_{\vec{\pi}}(a \otimes a' \otimes a'' \otimes a''').$$

With these definitions, if $w_1, w_2 \in W^{\otimes 6}$ are indicator tensors for 6-tuples of vertices, then $A[\tau^4_{\vec{\pi}}](w_1, w_2)$ counts the number of (possibly degenerate) steps of type $\vec{\pi}$ with specified attach tuples w_1 and w_2. In addition, since $A[\tau^4_{\vec{\pi}}]$ is a bilinear map $W^{\otimes 6} \times W^{\otimes 6} \to \mathbb{R}$, it corresponds to a square matrix. Also, (10) shows that $A[\tau^4_{\vec{\pi}}]$ is a symmetric matrix.

The ℓth matrix power of $A[\tau^4_{\vec{\pi}}]$ counts walks of length 2ℓ and type $\vec{\pi}$ by considering the form of matrix multiplication in (8). Indeed, the product $A[\tau^4_{\vec{\pi}}]^{\ell-1} A[\tau^4_{\vec{\pi}}]$ sums over the standard basis of $W^{\otimes 6}$, which can be considered as summing over the possible 6-vertex tuples used for the internal attach points of a walk of length $2(\ell - 1)$ and a step. Finally, the trace of $A[\tau^4_{\vec{\pi}}]^{\ell}$ then counts the number of walks of length 2ℓ with both attach tuples identified and these are exactly the circuits of length 2ℓ.

6.3 Products and powers of multilinear maps

In this section, we give the formal definitions of $\phi^{2^{t-1}}$ and $A[\phi^{2^{t-1}}]$ for a t-linear map ϕ needed for Proposition 8.

Definition. *Let $V_1, \dots, V_t$ be finite dimensional vector spaces over $\mathbb{R}$ and let $\phi, \psi : V_1 \times \dots \times V_t \to \mathbb{R}$ be t-linear maps. The* product *of ϕ and ψ, written $\phi * \psi$, is a $(t-1)$-linear map defined as follows. Let $u_1, \dots, u_{t-1}$ be vectors where $u_i \in V_i$. Let $\{b_1, \dots, b_{\dim(V_t)}\}$ be any* *orthonormal basis of* V_t*.*

$$\phi * \psi : (V_1 \otimes V_1) \times (V_2 \otimes V_2) \times \cdots \times (V_{t-1} \otimes V_{t-1}) \to \mathbb{R}$$
$$\phi * \psi(u_1 \otimes v_1, \ldots, u_{t-1} \otimes v_{t-1}) := \sum_{j=1}^{\dim(V_t)} \phi(u_1, \ldots, u_{t-1}, b_j)\psi(v_1, \ldots, v_{t-1}, b_j)$$

Extend the map $\phi * \psi$ *linearly to all of the domain to produce a* $(t-1)$*-linear map.*

Lemma 17 shows the above definition is well defined: the map is consistent when extending linearly and the map is the same for any choice of orthonormal basis $\{b_1, \ldots, b_{\dim(V_t)}\}$ of V_t.

Definition. *Let* $V_1, \ldots, V_t$ *be finite dimensional vector spaces over* $\mathbb{R}$ *and let* $\phi : V_1 \times \cdots \times V_t \to \mathbb{R}$ *be a* t*-linear map and let* s *be an integer* $0 \leq s \leq t-1$*. Define*

$$\phi^{2^s} : V_1^{\otimes 2^s} \times \cdots \times V_{t-s}^{\otimes 2^s} \to \mathbb{R}$$

where $\phi^{2^0} := \phi$ *and* $\phi^{2^s} := \phi^{2^{s-1}} * \phi^{2^{s-1}}$*.*

Note that we only define this for exponents which are powers of two because the product $*$ is only defined when the domains of the maps are the same. An expression like $\phi^3 = \phi*(\phi*\phi)$ does not make sense because ϕ and $\phi * \phi$ have different domains. This defines the power $\phi^{2^{t-1}}$, which is a linear map $V_1^{\otimes 2^{t-1}} \to \mathbb{R}$. In the next section, we will prove that $\tau_{\vec{\pi}}^{2^{t-1}}$ counts the number of labeled steps of type $\vec{\pi}$, similar to (10). As we saw in Section 6.2, we need to "untangle" the attach tuples of the input to $\tau_{\vec{\pi}}^{2^{t-1}}$.

Definition. *Let* $V_1, \ldots, V_t$ *be finite dimensional vector spaces over* $\mathbb{R}$ *and let* $\phi : V_1 \times \cdots \times V_t \to \mathbb{R}$ *be a* t*-linear map and define* $A[\phi^{2^{t-1}}]$ *to be the following square matrix/bilinear map. Let* $u_1, \ldots, u_{2^{t-2}}, v_1, \ldots, v_{2^{t-2}}$ *be vectors where* $u_i, v_i \in V_1$*.*

$$A[\phi^{2^{t-1}}] : V_1^{\otimes 2^{t-2}} \times V_1^{\otimes 2^{t-2}} \to \mathbb{R}$$
$$A[\phi^{2^{t-1}}](u_1 \otimes \cdots \otimes u_{2^{t-2}}, v_1 \otimes \ldots v_{2^{t-2}}) := \phi^{2^{t-1}}(u_1 \otimes v_1 \otimes u_2 \otimes v_2 \otimes \cdots \otimes u_{2^{t-2}} \otimes v_{2^{t-2}}).$$

Extend the map linearly to the entire domain to produce a bilinear map.

Lemma 17 from the appendix shows that the above definition is well defined, i.e. the map is consistent when extended linearly. By definition, $A[\phi^{2^{t-1}}]$ is a square real valued matrix. The next lemma, which is also proved in the appendix, shows that it is symmetric.

Lemma 11. *Let* $V_1, \ldots, V_t$ *be finite dimensional vector spaces over* $\mathbb{R}$*. If* $\phi : V_1 \times \cdots \times V_t \to \mathbb{R}$ *is a* t*-linear map, then* $A[\phi^{2^{t-1}}]$ *is a square symmetric real valued matrix.*

6.4 Proposition 8: counting walks and circuits

This section contains the proof of Proposition 8. All of the ideas in the proof are contained in Section 6.2, which showed that $\tau_{\vec{\pi}}$, $\tau_{\vec{\pi}}^2$, and $\tau_{\vec{\pi}}^4$ counted hypergraphs, in the sense that if indicator tensors of vertices are given as input, $\tau_{\vec{\pi}}^2$ and $\tau_{\vec{\pi}}^4$ count the number of some hypergraph where the vertices from the input are fixed and all other vertices are allowed to range over all of $V(H)$. It is straightforward from the definition to see that similarly for general k and $\vec{\pi}$ and for $0 \leq s \leq t-1$, $\tau_{\vec{\pi}}^{2^s}$ will count some hypergraph. Indeed, one can iteratively expand the definition of $\tau_{\vec{\pi}}^{2^s}$ to obtain an expression of the form

$$\tau_{\vec{\pi}}^{2^s}(\cdots) = \sum \cdots \sum \underbrace{\tau_{\vec{\pi}}(\cdots)\ \cdots\ \tau_{\vec{\pi}}(\cdots)}_{2^s}. \tag{11}$$

When the inputs to $\tau_{\vec{\pi}}^{2^s}$ are indicator tensors of vertices, the above expression counts some hypergraph since the sums are over bases and each basis vector can be thought of as the indicator tensor for some vertex tuple. Thus (11) counts some hypergraph, where the specific hypergraph depends on how the inputs and summation indices are arranged on the right hand side of (11). The proof in this section comes down to showing that the hypergraph which is counted by $\tau_{\vec{\pi}}^{2^{t-1}}$ is the step $S_{\vec{\pi}}$. We do this by induction by describing exactly the hypergraph counted by $\tau_{\vec{\pi}}^{2^s}$, which is the following hypergraph.

Definition. *For $\vec{\pi} = (1, \ldots, 1)$ with t parts, let $0 \leq s \leq t-1$ and define the hypergraph $D_{\vec{\pi},s}$ as follows. Let $A_1, \ldots, A_{t-s}$ be disjoint sets of size 2^s where elements are labeled by binary strings of length s and let $B_{t-s+1}, \ldots, B_t$ be disjoint sets of size 2^{s-1} where elements are labeled by binary strings of length $s-1$. The vertex set of $D_{\vec{\pi},s}$ is $A_1 \dot\cup \cdots \dot\cup A_{t-s} \dot\cup B_{t-s+1} \dot\cup \cdots \dot\cup B_t$. Make $a_1, \ldots, a_{t-s}, b_{t-s+1}, \ldots, b_t$ an edge of $D_{\vec{\pi},s}$ if $a_i \in A_i$, $b_j \in B_j$, the codes for $a_1, \ldots, a_{t-s}$ are all equal, and the code for b_{t-s+j} is equal to the code formed by removing the jth bit of the code for a_1.*

For a general $\vec{\pi} = (k_1, \ldots, k_t)$, start with $D_{(1,\ldots,1),s}$ and expand each vertex into the appropriate size; that is, a vertex in A_i is expanded into k_i vertices and each vertex in B_j is expanded into k_j vertices. In $D_{\vec{\pi},s}$, each vertex in A_i is labeled by a pair (c, z) where c is a bit string of length s and $z \in [k_i]$. We call z the expansion index *of the vertex.*

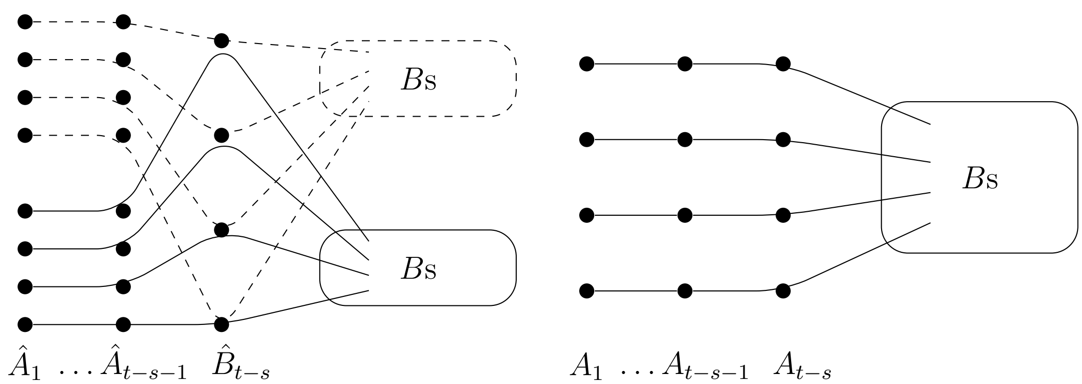
The hypergraph $D_{\vec{\pi},0}$ is a single edge and the hypergraph $D_{\vec{\pi},t-1}$ is by definition the step $S_{\vec{\pi}}$. The following lemma precisely formulates what we mean when we say that $\tau_{\vec{\pi}}^{2^s}$ counts the hypergraph $D_{\vec{\pi},s}$.

Lemma 12. *Let H be a k-uniform hypergraph, $\vec{\pi}$ a proper ordered partition of k with $\vec{\pi} = (k_1, \ldots, k_t)$, and let $0 \leq s \leq t-1$. Let W be the vector space over $\mathbb{R}$ of dimension $|V(H)|$ and let τ be the adjacency map of H. Let $A_1, \ldots, A_{t-s}, B_{t-s+1}, \ldots, B_t$ be the vertex sets in the definition of $D_{\vec{\pi},s}$ and let Δ be any map $A_1 \cup \cdots \cup A_{t-s} \to V(H)$. Then $\tau_{\vec{\pi}}^s$ counts the number of labeled, possibly degenerate copies of $D_{\vec{\pi},s}$ extending Δ as follows.*

Let $a_{i,1}, \ldots, a_{i,k_i 2^s}$ be the vertices of A_i ordered first lexicographically by bit code and then for equal codes ordered by expansion index. Let χ_i be the indicator tensor in $W^{\otimes k_i 2^s}$ for the vertex tuple $(\Delta(a_{i,1}), \ldots, \Delta(a_{i,k_i 2^s}))$. Then $\tau_{\vec{\pi}}^{2^s}(\chi_1, \ldots, \chi_{t-s})$ is the number of edge-preserving maps $V(D_{\vec{\pi},s}) \to V(H)$ which are consistent with Δ.

Proof. By induction on s. The base case is $s = 0$, where $D_{\vec{\pi},0}$ is a single edge, there are no B-type sets, and thus Δ is a map $V(D_{\vec{\pi},0}) \to V(H)$. The number of edge preserving maps extending Δ is either zero or one depending on if the image of Δ is an edge of H or not. But $\tau_{\vec{\pi}}(\chi_1, \ldots, \chi_t)$ equals zero or one depending on if the vertices defining the indicator tensors χ_i form an edge, exactly what is required.

Assume the lemma is true for s; we will prove it for $s+1$. Denote by $\hat{A}_1, \ldots, \hat{A}_{t-s-1}, \hat{B}_{t-s}, \ldots, \hat{B}_t$ the sets in the definition of $D_{\vec{\pi},s+1}$ and $A_1, \ldots, A_{t-s}, B_{t-s+1}, \ldots, B_t$ the sets in the definition of $D_{\vec{\pi},s}$. Let $\hat{\Delta}$ be a map $\hat{A}_1 \cup \cdots \cup \hat{A}_{t-s-1} \to V(H)$ and let $\hat{\chi}_1, \ldots, \hat{\chi}_{t-s-1}$ be the indicator tensors for the image of $\hat{\Delta}$ ordered as in the statement of the lemma. Since $\hat{\chi}_i$ is an indicator tensor in $W^{\otimes k_i 2^{s+1}}$, it is a simple tensor so $\hat{\chi}_i = \chi_i \otimes \chi'_i$ for $\chi_i, \chi'_i \in W^{\otimes k_i 2^s}$. Note that χ_i is the indicator tensor for the image under $\hat{\Delta}$ of the vertices of $D_{\vec{\pi},s+1}$ whose code starts with zero and χ'_i is the indicator tensor for the image under $\hat{\Delta}$ of the vertices whose code starts with a one, since the definition of $\hat{\chi}_i$ sorted the vertices in the image lexicographically.



(a) $D_{\vec{\pi},s+1}$ (b) $D_{\vec{\pi},s}$

$$\tau_{\vec{\pi}}^{2^{s+1}}(\chi_1 \otimes \chi'_1, \ldots, \chi_{t-s-1} \otimes \chi'_{t-s-1}) = \sum_{j=1}^{d} \tau_{\vec{\pi}}^{2^s}(\chi_1, \ldots, \chi_{t-s-1}, w_j)\tau_{\vec{\pi}}^{2^s}(\chi'_1, \ldots, \chi'_{t-s-1}, w_j) \tag{12}$$

Figure 8: The induction step of Lemma 12

Consider the expansion of the definition of $\tau_{\vec{\pi}}^{2^{s+1}}(\hat{\chi}_1, \ldots, \hat{\chi}_{t-s-1})$ shown in (12) in Figure 8; the tensors $\hat{\chi}_i$ are split into χ_i and χ'_i and we sum over the standard basis $\{w_1, \ldots, w_d\}$ of $W^{\otimes k_{t-s} 2^s}$, where $d = \dim(W^{\otimes k_{t-s} 2^s})$. This is a generalization of Section 6.2 where (9) was compared with Figure 6 and (10) was compared with Figure 7, where Figure 6 corresponds to Figure 8 *(b)* and Figure 7 corresponds to Figure 8 *(a)*. We can consider the tensor w_j in (12) to be the indicator tensor of a tuple of $k_{t-s}2^s$ vertices.

Definition. *We now describe two embeddings of $D_{\vec{\pi},s}$ into $D_{\vec{\pi},s+1}$. In Figure 8 (a), these two embeddings are the dotted and solid lines. Let $\Gamma_0 : V(D_{\vec{\pi},s}) \to V(D_{\vec{\pi},s+1})$ be the following* *injection. For $1 \le i \le t-s-1$ and $a \in A_i$, set $\Gamma_0(a)$ equal to the vertex in $\hat{A}_i$ whose code equals the code for a with a zero prepended to the code and the same expansion index. That is, a vertex in A_i with label $(1011, 4)$ is mapped to the vertex in $\hat{A}_i$ with label $(01011, 4)$. For $a \in A_{t-s}$, set $\Gamma_0(a)$ equal to the vertex in $\hat{B}_{t-s}$ which has the same label as a. For $t-s+1 \le j \le t$ and $b \in B_j$, set $\Gamma_0(b)$ equal to the vertex in $\hat{B}_j$ whose code equals the code for b with a zero prepended to the code and the same expansion index. In other words, Γ_0 adds a zero to the front of the codes except for vertices in A_{t-s} whose code does not change. Define $\Gamma_1 : V(D_{\vec{\pi},s}) \to V(D_{\vec{\pi},s+1})$ similarly except prepend a one instead of a zero. In Figure 8 (a), the dotted lines represent Γ_0 and the solid lines represent Γ_1.*

Claim: Γ_0 and Γ_1 are edge preserving injections and every edge in $D_{\vec{\pi},s+1}$ is in the image of Γ_0 or Γ_1 but not both.

Proof of Claim. Let E be an edge in $D_{\vec{\pi},s}$. For $1 \le i \le j \le t-s-1$ and $a_i \in A_i \cap E$ and $a_j \in A_j \cap E$, since E is an edge of $D_{\vec{\pi},s}$ the code for a_i equals the code for a_j. This implies that the codes for $\Gamma_0(a_i)$ and $\Gamma_0(a_j)$ are equal since both had a zero prepended. Now consider $b \in A_{t-s} \cap E$ which is mapped to $\hat{B}_{t-s}$. The conditions for $\Gamma_0(E)$ an edge of $D_{\vec{\pi},s+1}$ requires that the code for $\Gamma_0(b)$ equals the code formed by deleting the first bit of $\Gamma_0(a)$ where $a \in A_1 \cap E$. But the code for a equals the code for b since both are in A-type sets in $D_{\vec{\pi},s}$ and the map Γ_0 adds a zero to the front of the code for a and leaves the code for b alone. Thus the code for $\Gamma_0(b)$ equals the code formed by deleting the first bit of $\Gamma_0(a)$. Lastly, consider $b \in B_j \cap E$ for $t-s+1 \le j \le t$ and consider deleting the $(j+1)$-th bit of the code for $\Gamma_0(a)$. This is the same as deleting the j-th bit of a since $\Gamma_0(a)$ had a zero prepended. But deleting the jth bit of a equals the code for b, since $a, b \in E \in E(D_{\vec{\pi},s})$. Thus deleting the $(j+1)$th bit of $\Gamma_0(a)$ is the code for $\Gamma_0(b)$. We have now checked all the conditions, so $\Gamma_0(E)$ is an edge of $D_{\vec{\pi},s+1}$, i.e. Γ_0 is edge preserving. Γ_1 is edge preserving by the same argument. Finally, let E be an edge of $D_{\vec{\pi},s+1}$ and pick $a \in E \cap \hat{A}_1$. If the first bit of the code for a equals zero, then E is in the image of Γ_0 and if the first bit of the code for a equals one, then E is in the image of Γ_1. This concludes the proof of the claim. $\square$

This claim implies that any edge-preserving map extending $\hat{\Delta}$ is formed from two edge preserving maps $V(D_{\vec{\pi},s}) \to V(H)$ each extending the appropriate restriction of $\hat{\Delta}$. Start with $\hat{\Delta}$ and extend arbitrarily to a map $\Lambda : \hat{A}_1 \cup \cdots \cup \hat{A}_{t-s-1} \cup \hat{B}_{t-s} \to V(H)$. Next define Λ_0 and Λ_1 as maps $A_1 \cup \cdots \cup A_{t-s} \to V(H)$ such that $\Lambda_0 = \Lambda \circ \Gamma_0|_{\bar{A}}$ and $\Lambda_1 = \Lambda \circ \Gamma_1|_{\bar{A}}$, where $\bar{A} = A_1 \cup \cdots \cup A_{t-s}$ so $\Gamma_0|_{\bar{A}}$ is the map Γ_0 restricted to the A-type sets in $D_{\vec{\pi},s}$. By the claim, the number of edge-preserving maps extending $\hat{\Delta}$ equals the sum over Λ of the product of the number of edge-preserving maps extending Λ_0 and extending Λ_1. This is because any edge preserving map extending Λ can be composed with Γ_0 and Γ_1 to create edge preserving maps extending Λ_0 and Λ_1, and since Γ_0 and Γ_1 are injections covering all edges of $D_{\vec{\pi},s+1}$, this can be reversed. The last step in the proof is to show that this is exactly what (12) counts.

Let $\hat{b}_1, \ldots, \hat{b}_{k_{t-s}2^s}$ be the vertices of $\hat{B}_{t-s}$ listed first in lexicographic order of codes and then by expansion index. Let w be the indicator tensor in $W^{\otimes k_{t-s}2^s}$ for the vertex tuple

$(\Lambda(\hat{b}_1), \ldots, \Lambda(\hat{b}_{k_{t-s}2^s}))$. Note that as Λ ranges over all possible extensions of $\hat{\Delta}$, w ranges over the standard basis of $W^{\otimes k_{t-s}2^s}$. Now $\chi_1, \ldots, \chi_{t-s-1}, w$ are the indicator tensors representing the image of the map Λ_0, since as mentioned above, $\chi_1, \ldots, \chi_{t-s-1}$ are the indicator tensors for the image under $\hat{\Delta}$ of the vertices whose code stars with a zero. Similarly, $\chi'_1, \ldots, \chi'_{t-s-1}, w$ are the indicator tensors representing the image of the map Λ_1. Thus by induction, $\tau_{\vec{\pi}}^{2^s}(\chi_1, \ldots, \chi_{t-s-1}, w)$ is the number of edge-preserving maps extending Λ_0 and $\tau_{\vec{\pi}}^{2^s}(\chi'_1, \ldots, \chi'_{t-s-1}, w)$ is the number of edge preserving maps extending Λ_1. By the claim, this implies that the product

$$\tau_{\vec{\pi}}^{2^s}(\chi_1, \ldots, \chi_{t-s-1}, w)\tau_{\vec{\pi}}^{2^s}(\chi'_1, \ldots, \chi'_{t-s-1}, w)$$

counts the number of edge-preserving maps extending Λ. Thus (12) sums over the choices for Λ extending $\hat{\Delta}$ of the number of edge-preserving maps extending Λ. This sum is exactly the number of edge-preserving maps extending $\hat{\Delta}$, so the proof is complete. $\square$

Corollary 13. *Let H be a k-uniform hypergraph, $\vec{\pi}$ a proper ordered partition of k with $\vec{\pi} = (k_1, \ldots, k_t)$, and let $\ell \geq 2$ be an integer. Let W be the vector space over $\mathbb{R}$ of dimension $|V(H)|$ and let τ be the adjacency map of H. Let $a_1, \ldots, a_{k_1 2^{t-2}}, a'_1, \ldots, a'_{k_1 2^{t-2}}$ be (not necessarily distinct) vertices of H and let ξ and ξ' be the indicator tensors in $W^{k_1 2^{t-2}}$ for the tuples $(a_1, \ldots, a_{k_1 2^{t-2}})$ and $(a'_1, \ldots, a'_{k_1 2^{t-2}})$ respectively. Then $A[\tau_{\vec{\pi}}^{2^{t-1}}](\xi, \xi')$ is the number of labeled, possibly degenerate steps of type $\vec{\pi}$ in H with attach tuples $(a_1, \ldots, a_{k_1 2^{t-2}})$ and $(a'_1, \ldots, a'_{k_1 2^{t-2}})$. Also, $A[\tau_{\vec{\pi}}^{2^{t-1}}]^\ell(\xi, \xi')$ is the number of labeled walks of length 2ℓ and type $\vec{\pi}$ with attach tuples $(a_1, \ldots, a_{k_1 2^{t-2}})$ and $(a'_1, \ldots, a'_{k_1 2^{t-2}})$.*

Proof. The proof is by induction on ℓ. First, consider the base case of $\ell = 1$, where the path of length two and type $\vec{\pi}$ is the step of type $\vec{\pi}$. Let A be the vertex set from the definition of the step $S_{\vec{\pi}}$. Define a mapping $\Delta : A \to V(H)$ by mapping the attach tuples of $S_{\vec{\pi}}$ to the tuples $(a_1, \ldots, a_{k_1 2^{t-2}})$ and $(a'_1, \ldots, a'_{k_1 2^{t-2}})$ in $V(H)$. By definition, the first attach tuple of $S_{\vec{\pi}}$ is the vertices ending with a zero and listed in lexicographic order and the second attach tuple of $S_{\vec{\pi}}$ is the vertices ending with a one and listed in lexicographic order. This implies that the indicator tensor χ_1 from the statement of Lemma 12 is the indicator tensor in $W^{\otimes k_1 2^{t-1}}$ for the tuple $(a_1, \ldots, a_{k_1}, a'_1, \ldots, a'_{k_1}, a_{k_1+1}, \ldots, a_{2k_1}, a'_{k_1+1}, \ldots, a'_{2k_1}, \ldots, a_{k_1 2^{t-3}+1}, \ldots, a_{k_1 2^{t-2}}, a'_{k_1 2^{t-3}+1}, \ldots, a'_{k_1 2^{t-2}})$, since each attach tuple is in lexicographic order but the last bit is zero or one so the full ordering alternates between attach tuples. By the definition of $A[\tau_{\vec{\pi}}^{2^{t-1}}]$ and the indicator tensors ξ, ξ', χ_1, $A[\tau_{\vec{\pi}}^{2^{t-1}}](\xi, \xi') = \tau_{\vec{\pi}}^{2^{t-1}}(\chi_1)$. Thus Lemma 12 applied with $s = t-1$ shows that the number of edge-preserving maps extending Δ is $A[\tau_{\vec{\pi}}^{2^{t-1}}](\xi, \xi')$, but by the definition of Δ, this is exactly the number of labeled, possibly degenerate steps of type $\vec{\pi}$ with attach tuples $(a_1, \ldots, a_{k_1 2^{t-2}})$ and $(a'_1, \ldots, a'_{k_1 2^{t-2}})$.

Next assume that the corollary is true for ℓ; we will show that it is true for $\ell + 1$. Using the form of matrix multiplication in (8), let $\{d_1, \ldots, d_{\dim(W^{\otimes k_1 2^{t-2}})}\}$ be the standard basis of $W^{\otimes k_1 2^{t-2}}$ so

$$A[\tau_{\vec{\pi}}^{2^{t-1}}]^{\ell+1}(\xi, \xi') = \sum_{i=1}^{\dim(W^{\otimes k_1 2^{t-2}})} A[\tau_{\vec{\pi}}^{2^{t-1}}]^\ell(\xi, d_i) A[\tau_{\vec{\pi}}^{2^{t-1}}](d_i, \xi'). \tag{13}$$

Each standard basis vector d_i can be thought of as a $k_1 2^{t-2}$-tuple of vertices which corresponds to one of the two attach tuples. Thus (13) sums over the internal attach tuple for a walk of length 2ℓ and $S_{\vec{\pi}}$. □

Proof of Proposition 8. Since $A[\tau_{\vec{\pi}}^{2^{t-1}}]^\ell$ counts the number of walks of length 2ℓ, the trace counts circuits. If $\{d_1, \dots, d_{\dim(W^{\otimes k_1 2^{t-2}})}\}$ is any orthonormal basis of $W^{\otimes k_1 2^{t-2}}$, the trace of the matrix $A[\tau_{\vec{\pi}}^{2^{t-1}}]^\ell$ is

$$\mathrm{Tr}\left[A[\tau_{\vec{\pi}}^{2^{t-1}}]^\ell\right] = \sum_{i=1}^{\dim(W^{\otimes k_1 2^{t-2}})} A[\tau_{\vec{\pi}}^{2^{t-1}}]^\ell(d_i, d_i).$$

If $\{d_1, \dots, d_{\dim(W^{\otimes k_1 2^{t-2}})}\}$ is the standard basis, each d_i corresponds to a tuple of $k_1 2^{t-2}$ vertices, so the above expression is the number of walks of type $\vec{\pi}$ with both attach tuples equal to d_i. □

7 Concluding Remarks

- A beautiful conjecture of Sidorenko [43] asks about the minimum number of copies of a bipartite graph H which must be contained in another graph G. Let $h_H(G)$ denote the number of labeled copies of H in G, i.e. the number of edge-preserving maps from $V(H)$ to $V(G)$. Let $t_H(G) = h_H(G)/|V(G)|^{|V(H)|}$ be the normalization of $h_H(G)$, which is the fraction of mappings $f : V(H) \to V(G)$ which are edge preserving. Sidorenko's Conjecture is that for every bipartite graph H with m edges and every graph G, $t_H(G) \geq t_{K_2}(G)^m$. Sidorenko's Conjecture is known to be true in a small number of cases [16, 29, 38, 43].

  For k-uniform hypergraphs with $k \geq 3$, Sidorenko [43] gave a construction (the loose triangle) for which the generalization of the conjecture to k-uniform hypergraphs fails. Nevertheless, Sidorenko [44] proved that the generalization of the conjecture holds for so-called "hypergraph trees": hypergraphs where the edges can be ordered so that there is a degree one vertex in the union of the first i edges in this ordering for every i. In addition, standard arguments can be used to show that the conjecture holds for complete k-uniform, k-partite hypergraphs. This leads to the following problem.

  Problem 14. *For which k-partite, k-uniform hypergraphs H with m edges is it true that every k-uniform hypergraph G has $t_H(G) \geq t_{K_k}(G)^m$?*

  An easy consequence of the hypergraph eigenvalue techniques used to prove Theorem 3 is that this is true for all cycles of type π and length 4ℓ, for any π and any ℓ.

  Theorem 15. *For every $k \geq 2$, every proper partition π of k, every $\ell \geq 1$, and every k-uniform hypergraph G,*

  $$t_{C_{\pi,4\ell}}(G) \geq t_{K_k}(G)^{|E(C_{\pi,4\ell})|}.$$

- The eigenvalue techniques developed in this paper lead to a polynomial-time algorithm to detect hypergraph quasirandomness. Our algorithm works by counting cycles and improves the algorithm of Hán, Person, and Schacht [28] by counting longer cycles and working with general $\vec{\pi}$. The analysis that the algorithm is correct relies on eigenvalues and is similar to the proof that $\texttt{Cycle}_{4\ell}[\pi] \Rightarrow \texttt{Eig}[\pi]$ from Section 6.1. Our algorithm leads to a strong refutation algorithm for random k-SAT, one of the most studied NP-complete problems. A detailed analysis of our algorithm appears online [35].

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A Algebraic properties of multilinear maps

In this section we prove several algebraic facts about multilinear maps, including Proposition 9 and Lemmas 10 and 11. Throughout this section, V and V_i are finite dimensional vector spaces over $\mathbb{R}$. Also in this section we make no distinction between bilinear maps and matrices, using whichever formulation is convenient. We will use a symbol $\cdot$ to denote the input to a linear map; for example, if $\phi : V_1 \times V_2 \times V_3 \to \mathbb{R}$ is a trilinear map and $x_1 \in V_1$ and $x_2 \in V_2$, then by the expression $\phi(x_1, x_2, \cdot)$ we mean the linear map from V_3 to $\mathbb{R}$ which takes a vector $x_3 \in V_3$ to $\phi(x_1, x_2, x_3)$. Lastly, we use several basic facts about tensors, all of which follow from the fact that we defined the tensor product of V and W to be the vector space over $\mathbb{R}$ of dimension $\dim(V)\dim(W)$. For example, if x and y are unit length, then $x \otimes y$ is also unit length.

Lemma 16. *Let $\phi : V \to \mathbb{R}$ be a linear map. There exists a vector v such that $\phi = \langle v, \cdot \rangle$.*

Proof. v is the vector dual to ϕ in the dual of the vector space V. Alternatively, let the ith coordinate of v be $\phi(e_i)$, since then for any x,

$$\phi(x) = \phi\left(\sum \langle x, e_i \rangle e_i\right) = \sum \langle x, e_i \rangle \phi(e_i) = \sum \langle x, e_i \rangle \langle v, e_i \rangle = \langle x, v \rangle .$$

□

Lemma 17. *Let $\phi, \psi : V_1 \times \cdots \times V_t \to \mathbb{R}$ be t-linear maps. The maps $\phi * \psi$ and $A[\phi^{2^{t-1}}]$ are well defined. Also, $\phi * \psi$ is basis independent in the sense that the definition of $\phi * \psi$ is independent of the choice of orthonormal basis $b_1, \ldots, b_t$ of V_t.*

Proof. First, extending the definitions of $\phi * \psi$ and $A[\phi^{2^{t-1}}]$ linearly to the entire domain (non-simple tensors) is well defined, since ϕ and ψ are linear. That is, write each u_i and v_i in terms of some orthonormal basis and expand each tensor in $V_i \otimes V_i$ also in terms of this basis. The linearity of ϕ and ψ then shows that the definitions of $\phi * \psi$ and $A[\phi^{2^{t-1}}]$ are well defined and linear. To see basis independence of $\phi * \psi$, by Lemma 16 the linear map $\phi(u_1, \ldots, u_{t-1}, \cdot) : V_t \to \mathbb{R}$ equals $\langle u', \cdot \rangle$ for some vector u'. Similarly, $\psi(v_1, \ldots, v_t, \cdot)$ equals $\langle v', \cdot \rangle$ for some vector v'. Then

$$(\phi * \psi)(u_1 \otimes v_1, \ldots, u_{t-1} \otimes v_{t-1}) = \sum_{i=1}^{\dim(V_t)} \langle u', b_i \rangle \langle v', b_i \rangle = \langle u', v' \rangle .$$

The last equality is valid for any orthonormal basis, since the dot product of u' and v' sums the product of the ith coordinate of u' in the basis $\{b_1, \ldots, b_{\dim(V_t)}\}$ with the ith coordinate of v' in the basis $\{b_1, \ldots, b_{\dim(V_t)}\}$. $\square$

Definition. *For $s \geq 0$ and V a finite dimensional vector space over $\mathbb{R}$, define the vector space isomorphism $\Gamma_{V,s} : V^{\otimes 2^s} \to V^{\otimes 2^s}$ as follows. If $s = 0$, define $\Gamma_{V,0}$ to be the identity map. If $s \geq 1$, let $\{b_1, \ldots, b_{\dim(V)}\}$ be any orthonormal basis of V and define for all $(i_1, \ldots, i_{2^{s-1}}, j_1, \ldots, j_{2^{s-1}}) \in [\dim(V)]^{2^s}$,*

$$\Gamma_{V,s}(b_{i_1} \otimes b_{j_1} \otimes \cdots \otimes b_{i_{2^{s-1}}} \otimes b_{j_{2^{s-1}}}) = b_{j_1} \otimes b_{i_1} \otimes \cdots \otimes b_{j_{2^{s-1}}} \otimes b_{i_{2^{s-1}}}. \tag{14}$$

Extend $\Gamma_{V,s}$ linearly to all of $V^{\otimes 2^s}$.

Remarks. $\Gamma_{V,s}$ is a vector space isomorphism since it restricts to a bijection of an orthonormal basis to itself. Also, it is easy to see that $\Gamma_{V,s}$ is well defined and independent of the choice of orthonormal basis, since each b_i can be written as a linear combination of an orthonormal basis $\{b'_1, \ldots, b'_{\dim(V)}\}$ and (14) can be expanded using linearity. For notational convenience, we will usually drop the subscript V and write Γ_s for $\Gamma_{V,s}$.

Lemma 18. *Let $\phi : V_1 \times \cdots \times V_t \to \mathbb{R}$ be a t-linear map, let $0 \leq s \leq t-1$, and let $x_1 \in V_1^{\otimes 2^s}, \ldots, x_{t-s} \in V_{t-s}^{\otimes 2^s}$. Then*

$$\phi^{2^s}(x_1, \ldots, x_{t-s}) = \phi^{2^s}(\Gamma_s(x_1), \ldots, \Gamma_s(x_{t-s})).$$

Proof. By induction on s. The base case is $s = 0$ where Γ_0 is the identity map. Expand the definition of $\phi^{2^{s+1}}$ and use induction to obtain

$$\phi^{2^{s+1}}(x_1 \otimes y_1, \ldots, x_{t-s-1} \otimes y_{t-s-1}) = \sum_{j=1}^{\dim(V_{t-s}^{\otimes 2^s})} \phi^{2^s}(x_1, \ldots, x_{t-s-1}, b_j)\phi^{2^s}(y_1, \ldots, y_{t-s-1}, b_j)$$

$$= \sum_{j=1}^{\dim(V_{t-s}^{\otimes 2^s})} \phi^{2^s}\big(\Gamma_s(x_1), \ldots, \Gamma_s(x_{t-s-1}), \Gamma_s(b_j)\big)\phi^{2^s}\big(\Gamma_s(y_1), \ldots, \Gamma_s(y_{t-s-1}), \Gamma_s(b_j)\big).$$

But since Γ_s is a vector space isomorphism, $\{\Gamma_s(b_1), \ldots, \Gamma_s(b_{\dim(V_{t-s}^{\otimes 2^s})})\}$ is an orthonormal basis of $V_{t-s}^{\otimes 2^s}$. Thus Lemma 17 shows that

$$\sum_{j=1}^{\dim(V_{t-s}^{\otimes 2^s})} \phi^{2^s}\big(\Gamma_s(x_1), \ldots, \Gamma_s(x_{t-s-1}), \Gamma_s(b_j)\big)\phi^{2^s}\big(\Gamma_s(y_1), \ldots, \Gamma_s(y_{t-s-1}), \Gamma_s(b_j)\big)$$
$$= \phi^{2^{s+1}}\big(\Gamma_s(x_1) \otimes \Gamma_s(y_1), \ldots, \Gamma_s(x_{t-s-1}) \otimes \Gamma_s(y_{t-s-1})\big)$$

Finally, $\Gamma_s(x_i) \otimes \Gamma_s(y_i) = \Gamma_{s+1}(x_i \otimes y_i)$ (write x_i and y_i as linear combinations, expand $\Gamma_{s+1}(x_i \otimes y_i)$ using linearity, and apply (14)). Thus $\phi^{2^{s+1}}(x_1 \otimes y_1, \ldots, x_{t-s-1} \otimes y_{t-s-1}) = \phi^{2^{s+1}}(\Gamma_{s+1}(x_1 \otimes y_1), \ldots, \Gamma_{s+1}(x_{t-s-1} \otimes y_{t-s-1}))$, completing the proof. $\square$

Proof of Lemma 11. Let $\phi : V_1 \times \cdots \times V_t \to \mathbb{R}$ be a t-linear map. $A[\phi^{2^{t-1}}]$ is a bilinear map from $V_1^{\otimes 2^{t-2}} \times V_1^{\otimes 2^{t-2}} \to \mathbb{R}$ and so is a square matrix of dimension $\dim(V_1)^{2^{t-2}}$. Lemma 18 shows that $A[\phi^{2^{t-1}}]$ is a symmetric matrix, since

$$\begin{aligned} A[\phi^{2^{t-1}}](x_1 \otimes \cdots \otimes x_{2^{t-2}}, y_1 \otimes \cdots \otimes y_{2^{t-2}}) &= \phi^{2^{t-1}}(x_1 \otimes y_1 \otimes \cdots \otimes x_{2^{t-2}} \otimes y_{2^{t-2}}) \\ &= \phi^{2^{t-1}}(\Gamma(x_1 \otimes y_1 \otimes \cdots \otimes x_{2^{t-2}} \otimes y_{2^{t-2}})) \\ &= \phi^{2^{t-1}}(y_1 \otimes x_1 \otimes \cdots \otimes y_{2^{t-2}} \otimes x_{2^{t-2}}) \\ &= A[\phi^{2^{t-1}}](y_1 \otimes \cdots \otimes y_{2^{t-2}}, x_1 \otimes \cdots \otimes x_{2^{t-2}}). \end{aligned}$$

The above equation is valid for all $x_i, y_i \in V_1$, in particular for all basis elements of V_1 which implies that $A[\phi^{2^{t-1}}](w, z) = A[\phi^{2^{t-1}}](z, w)$ for all basis vectors w, z of $V_1^{\otimes 2^{t-2}}$. Thus $A[\phi^{2^{t-1}}]$ is a square symmetric real-valued matrix. $\square$

Lemma 19. *Let $\phi : V_1 \times \cdots \times V_t \to \mathbb{R}$ be a t-linear map and let $x_1 \in V_1, \ldots, x_t \in V_t$ be unit length vectors. Then*

$$|\phi(x_1, \ldots, x_t)|^2 \le \left|\phi^2(x_1 \otimes x_1, \ldots, x_{t-1} \otimes x_{t-1})\right|.$$

Proof. Consider the linear map $\phi(x_1, \ldots, x_{t-1}, \cdot)$ which is a linear map from V_t to $\mathbb{R}$. By Lemma 16, there exists a vector $w \in V_t$ such that $\phi(x_1, \ldots, x_{t-1}, \cdot) = \langle w, \cdot \rangle$. Now expand out the definition of ϕ^2:

$$\phi^2(x_1 \otimes x_1, \ldots, x_{t-1} \otimes x_{t-1}) = \sum_j |\phi(x_1, \ldots, x_{t-1}, b_j)|^2 = \sum_j |\langle w, b_j \rangle|^2 = \langle w, w \rangle$$

where the last equality is because $\{b_j\}$ is an orthonormal basis of V_t. Since $\|w\| = \sqrt{\langle w, w \rangle}$,

$$\left|\phi^2(x_1 \otimes x_1, \ldots, x_{t-1} \otimes x_{t-1})\right| = |\langle w, w \rangle| = \left|\left\langle w, \frac{w}{\|w\|} \right\rangle\right|^2.$$

But since x_t is unit length and $\langle w, \cdot \rangle$ is maximized over the unit ball at vectors parallel to w (so maximized at $w / \|w\|$), $\left|\left\langle w, \frac{w}{\|w\|} \right\rangle\right| \geq |\langle w, x_t \rangle|$. Thus

$$\left|\phi^2(x_1 \otimes x_1, \ldots, x_{t-1} \otimes x_{t-1})\right| = \left|\left\langle w, \frac{w}{\|w\|} \right\rangle\right|^2 \geq |\langle w, x_t \rangle|^2 = |\phi(x_1, \ldots, x_t)|^2 .$$

The last equality used the definition of w, that $\phi(x_1, \ldots, x_{t-1}, \cdot) = \langle w, \cdot \rangle$. ☐

Lemma 20. *Let $\phi : V_1 \times \cdots \times V_t \to \mathbb{R}$ be a t-linear map and let $x_1 \in V_1, \ldots, x_t \in V_t$ be unit length vectors. Then for $0 \leq s \leq t-1$,*

$$|\phi(x_1, \ldots, x_t)|^{2^s} \leq \left|\phi^{2^s}(\underbrace{x_1 \otimes \cdots \otimes x_1}_{2^s}, \ldots, \underbrace{x_{t-s} \otimes \cdots \otimes x_{t-s}}_{2^s})\right|$$

which implies that

$$|\phi(x_1, \ldots, x_t)|^{2^{t-1}} \leq \left|A[\phi^{2^{t-1}}](\underbrace{x_1 \otimes \cdots \otimes x_1}_{2^{t-2}}, \underbrace{x_1 \otimes \cdots \otimes x_1}_{2^{t-2}})\right| .$$

Proof. By induction on s. The base case is $s = 0$ where both sides are equal and the induction step follows from Lemma 19. By definition of $A[\phi^{2^{t-1}}]$,

$$\left|A[\phi^{2^{t-1}}](\underbrace{x_1 \otimes \cdots \otimes x_1}_{2^{t-2}}, \underbrace{x_1 \otimes \cdots \otimes x_1}_{2^{t-2}})\right| = \left|\phi^{2^{t-1}}(\underbrace{x_1 \otimes \cdots \otimes x_1}_{2^{t-1}})\right| ,$$

completing the proof. ☐

Proof of Lemma 10. Pick $x_1, \ldots, x_t$ unit length vectors to maximize ϕ, so $\phi(x_1, \ldots, x_t) = \|\phi\|$. Then Lemma 20 shows that

$$\|\phi\|^{2^{t-1}} = |\phi(x_1, \ldots, x_t)|^{2^{t-1}} \leq \left|A[\phi^{2^{t-1}}](\underbrace{x_1 \otimes \cdots \otimes x_1}_{2^{t-2}}, \underbrace{x_1 \otimes \cdots \otimes x_1}_{2^{t-2}})\right|$$

Since $x_1 \otimes \cdots \otimes x_1$ is unit length, the above expression is upper bounded by the spectral norm of $A[\phi^{2^{t-1}}]$. ☐

Lemma 21. *Let $\{M_r\}_{r \to \infty}$ be a sequence of square symmetric real-valued matrices with dimension going to infinity where $\lambda_2(M_r) = o(\lambda_1(M_r))$. Let u_r be a unit length eigenvector corresponding to the largest eigenvalue in absolute value of M_r. If $\{x_r\}$ is a sequence of unit length vectors such that $\left|x_r^T M_r x_r\right| = (1 + o(1))\lambda_1(M_r)$, then*

$$\|u_r - x_r\| = o(1).$$

Consequently, for any unit length sequence $\{y_r\}$ where each y_r is perpendicular to x_r,

$$\left|y_r^T M_r y_r\right| = o(\lambda_1(M_r)).$$

Proof. Throughout this proof, the subscript r is dropped; all terms $o(\cdot)$ should be interpreted as $r \to \infty$. This exact statement was proved by Chung, Graham, and Wilson [15], although they don't clearly state it as such. We give a proof here for completeness using slightly different language but the same proof idea: if x projected onto $u^\perp$ is too big then the second largest eigenvalue is too big. Write $x = \alpha v + \beta u$ where v is a unit length vector perpendicular to u and $\alpha, \beta \in \mathbb{C}$ and $\alpha^2 + \beta^2 = 1$ (since u is an eigenvector it might have complex entries). Let $\phi(x, y) = x^T M y$ be the bilinear map corresponding to M. Since $u^T M v = \lambda_1 u^T v = \lambda_1 \langle u, v \rangle = 0$, we have $\phi(u, v) = 0$. This implies that

$$\begin{aligned}\phi(x, x) &= \phi(\alpha v + \beta u, \alpha v + \beta u) = \alpha^2 \phi(v, v) + \beta^2 \phi(u, u) + 2\alpha\beta\phi(u, v)\\ &= \alpha^2 \phi(v, v) + \beta^2 \phi(u, u).\end{aligned}$$

The second largest eigenvalue of M is the largest eigenvalue of $M - \lambda_1(M)uu^T$ which is the spectral norm of $M - \lambda_1(M)uu^T$. Thus

$$|\phi(v, v)| = |v^T M v| = |v^T (M - \lambda_1(M)uu^T)v| \le \lambda_2(M). \tag{15}$$

Using that $\phi(u, u) = \lambda_1(M)$ and the triangle inequality, we obtain

$$|\phi(x, x)| \le \alpha^2 \lambda_2(M) + \beta^2 \lambda_1(M). \tag{16}$$

Since $\alpha^2 + \beta^2 = 1$, $|\alpha|$ and $|\beta|$ are between zero and one. Combining this with (16) and $|\phi(x, x)| = (1 + o(1))\lambda_1(M)$ and $\lambda_2(M) = o(\lambda_1(M))$, we must have $|\beta| = 1 + o(1)$ which in turn implies that $|\alpha| = o(1)$. Consequently,

$$\|u - x\|^2 = \langle u - x, u - x \rangle = \langle u, u \rangle + \langle x, x \rangle - 2 \langle u, x \rangle = 2 - 2\beta = o(1).$$

Now consider some y perpendicular to x and similarly to the above, write $y = \gamma w + \delta u$ for some unit length vector w perpendicular to u and $\gamma, \delta \in \mathbb{C}$ with $\gamma^2 + \delta^2 = 1$. Then

$$\phi(y, y) = \phi(\gamma w + \delta u, \gamma w + \delta u) = \gamma^2 \phi(w, w) + \delta^2 \phi(u, u)$$

and as in (15), we have $|\phi(w, w)| \le \lambda_2(M)$. Thus

$$|\phi(y, y)| \le \gamma^2 \lambda_2(M) + \delta^2 \lambda_1(M).$$

We want to conclude that the above expression is $o(\lambda_1(M))$. Since $\lambda_2(M) = o(\lambda_1(M))$, we must prove that $|\delta| = o(1)$ to complete the proof.

$$\delta = \langle y, u \rangle = \left\langle y, \frac{x - \alpha v}{\beta} \right\rangle = \frac{1}{\beta}\big(\langle y, x \rangle - \alpha \langle y, v \rangle \big) = \frac{-\alpha \langle y, v \rangle}{\beta}.$$

But $|\alpha| = o(1)$, $|\beta| = 1 + o(1)$, and $\|y\| = \|v\| = 1$ so $|\delta| = o(1)$ as required. □

Lemma 22. *Let $J : V_1 \times \cdots \times V_t \to \mathbb{R}$ be the all-ones map and let $\vec{1}_i$ be the all-ones vector in V_i. Then for all $x_1, \ldots, x_t$ with $x_i \in V_i$,*

$$J(x_1, \ldots, x_t) = \left\langle \vec{1}_1, x_1 \right\rangle \cdots \left\langle \vec{1}_t, x_t \right\rangle. \tag{17}$$

Proof. If $x_1, \ldots, x_t$ are standard basis vectors, then the left and right hand side of (17) are the same. By linearity, (17) is then the same for all $x_1, \ldots, x_t$. $\square$

Proof of Proposition 9. Again throughout this proof, the subscript r is dropped; all terms $o(\cdot)$ should be interpreted as $r \to \infty$. Let $\hat{1}$ denote the all-ones vector scaled to unit length in the appropriate vector space. Pick an ordering $\vec{\pi} = (k_1, \ldots, k_t)$ of π. The definition of spectral norm is independent of the choice of the ordering for the entries of $\vec{\pi}$, so $\|\psi_{\vec{\pi}} - qJ_{\vec{\pi}}\|$ is the same for all orderings. Let $w_1, \ldots, w_t$ be unit length vectors where $(\psi_{\vec{\pi}} - qJ_{\vec{\pi}})(w_1, \ldots, w_t) = \|\psi_{\vec{\pi}} - qJ_{\vec{\pi}}\|$ and write $w_i = \alpha_i y_i + \beta_i \hat{1}$ where y_i is a unit length vector perpendicular to the all-ones vector and $\alpha_i, \beta_i \in \mathbb{R}$ with $\alpha_i^2 + \beta_i^2 = 1$. Then

$$\begin{aligned}\|\psi_{\vec{\pi}} - qJ_{\vec{\pi}}\| &= (\psi_{\vec{\pi}} - qJ_{\vec{\pi}})(w_1, \ldots, w_t) = (\psi_{\vec{\pi}} - qJ_{\vec{\pi}})(\alpha_1 y_1 + \beta_1 \hat{1}, \ldots, \alpha_t y_t + \beta_t \hat{1}) \\ &= \psi_{\vec{\pi}}(\alpha_1 y_1 + \beta_1 \hat{1}, \ldots, \alpha_t y_t + \beta_t \hat{1}) - q \dim(V_r)^{k/2} \prod \beta_i. \end{aligned} \tag{18}$$

The last equality used that y_i is perpendicular to $\hat{1}$, so Lemma 22 implies that if y_i appears as input to $J_{\vec{\pi}}$ then the outcome is zero no matter what the other vectors are. Thus the only non-zero term involving $J_{\vec{\pi}}$ is $J_{\vec{\pi}}(\hat{1}, \ldots, \hat{1}) = \dim(V_r)^{k/2}$. Note that $\psi(\hat{1}, \ldots, \hat{1}) = \psi_{\vec{\pi}}(\hat{1}, \ldots, \hat{1})$ since the all-ones vector scaled to unit length in $V^{\otimes k_i}$ is the tensor product of the all-ones vector scaled to unit length in V. Inserting $q = \dim(V_r)^{-k/2} \psi_{\vec{\pi}}(\hat{1}, \ldots, \hat{1})$ in (18), we obtain

$$\|\psi_{\vec{\pi}} - qJ_{\vec{\pi}}\| = \psi_{\vec{\pi}}(\alpha_1 y_1 + \beta_1 \hat{1}, \ldots, \alpha_t y_t + \beta_t \hat{1}) - \left(\prod \beta_i\right) \psi_{\vec{\pi}}(\hat{1}, \ldots, \hat{1}). \tag{19}$$

Now consider expanding $\psi_{\vec{\pi}}$ in (19) using linearity; the term $(\prod \beta_i)\psi_{\vec{\pi}}(\hat{1}, \ldots, \hat{1})$ cancels, so all terms include at least one y_i. We claim that each of these terms is small; the following claim finishes the proof, since $\|\psi_{\vec{\pi}} - qJ_{\vec{\pi}}\|$ is the sum of terms each of which $o(\psi(\hat{1}, \ldots, \hat{1}))$.

Claim: If $z_1, \ldots, z_{i-1}, z_{i+1}, \ldots, z_t$ are unit length vectors, then

$$|\psi_{\vec{\pi}}(z_1, \ldots, z_{i-1}, y_i, z_{i+1}, \ldots, z_t)| = o(\psi(\hat{1}, \ldots, \hat{1})).$$

Proof. Change the ordering of $\vec{\pi}$ to an ordering $\vec{\pi}'$ that differs from $\vec{\pi}$ by swapping 1 and i. Since ψ is symmetric,

$$\psi_{\vec{\pi}}(z_1, \ldots, z_{i-1}, y_i, z_{i+1}, \ldots, z_t) = \psi_{\vec{\pi}'}(y_i, z_2, \ldots, z_{i-1}, z_1, z_{i+1}, \ldots, z_t). \tag{20}$$

Therefore proving the claim comes down to bounding $\psi_{\vec{\pi}'}(y_i, z_2, \ldots, z_{i-1}, z_1, z_{i+1}, \ldots, z_t)$, which is a combination of Lemma 20 and Lemma 21 as follows. For the remainder of this proof, denote by A the matrix $A[\psi_{\vec{\pi}'}^{2^{t-1}}]$. By assumption, we have $\lambda_2(A) = o(\lambda_1(A))$ so Lemma 21 can be applied to the matrix sequence A. Next we would like to show that we can use $\hat{1}$ for x in the statement of Lemma 21; i.e. that $A(\hat{1}, \hat{1}) = (1 + o(1))\lambda_1(A)$. By Lemma 20 and the assumption $\lambda_1(A) = (1 + o(1))\psi(\hat{1}, \ldots, \hat{1})^{2^{t-1}}$, we have

$$\left|\psi_{\vec{\pi}'}(\hat{1}, \ldots, \hat{1})\right|^{2^{t-1}} \le \left|A(\hat{1}, \hat{1})\right| \le \lambda_1(A) = (1 + o(1))\psi\left(\hat{1}, \ldots, \hat{1}\right)^{2^{t-1}}.$$

Using the definition of $\psi_{\vec{\pi}'}$, we have $\psi_{\vec{\pi}'}(\hat{1}, \ldots, \hat{1}) = \psi(\hat{1}, \ldots, \hat{1})$, which implies asymtotic equality through the above equation. In particular, $|A(\hat{1}, \hat{1})| = (1 + o(1))\lambda_1(A)$ which is the condition in Lemma 21 for $x = \hat{1}$. Lastly, to apply Lemma 21 we need a vector y perpendicular to $\hat{1}$. The vector $y_i \otimes \cdots \otimes y_i \in V^{\otimes k_i 2^{t-2}}$ is pependicular to $\hat{1}$ (in $V^{\otimes k_i 2^{t-2}}$) since y_i itself is perpendicular to $\hat{1}$ (in $V^{\otimes k_i}$). Thus Lemma 21 implies that

$$\left| A(\underbrace{y_i \otimes \cdots \otimes y_i}_{2^{t-2}}, \underbrace{y_i \otimes \cdots \otimes y_i}_{2^{t-2}}) \right| = o(\lambda_1(A)). \tag{21}$$

Using Lemma 20 again shows that

$$|\psi_{\vec{\pi}'}(y_i, z_2, \ldots, z_{i-1}, z_1, z_{i+1}, \ldots, z_t)|^{2^{t-1}} \leq \left| A(\underbrace{y_i \otimes \cdots \otimes y_i}_{2^{t-2}}, \underbrace{y_i \otimes \cdots \otimes y_i}_{2^{t-2}}) \right|.$$

Combining this equation with (20) and (21) shows that $|\psi_{\vec{\pi}}(z_1, \ldots, z_{i-1}, y_i, z_{i+1}, \ldots, z_t)|^{2^{t-1}} = o(\lambda_1(A))$. By assumption, $\lambda_1(A) = (1 + o(1))\psi(\hat{1}, \ldots, \hat{1})^{2^{t-1}}$, completing the proof of the claim.

□